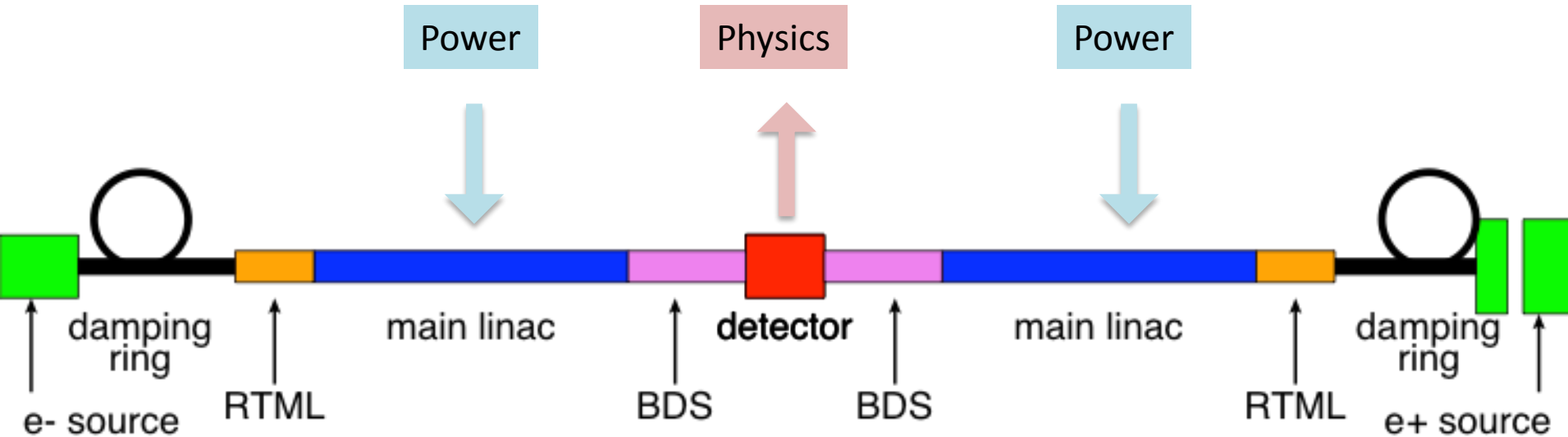


Beam-beam Effects in Linear Colliders

Daniel Schulte

Generic Linear Collider



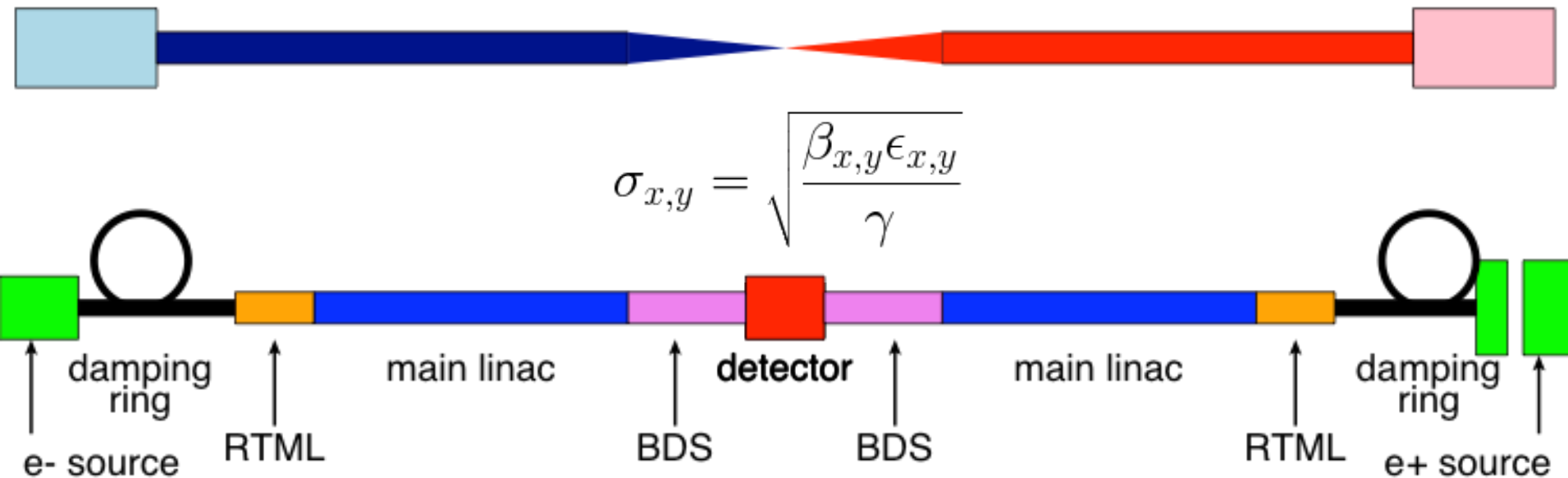
Can reach high electron-positron centre-of-mass energies

- almost no synchrotron radiation

Single pass, hence two main challenges

- gradient
- luminosity

Generic Linear Collider

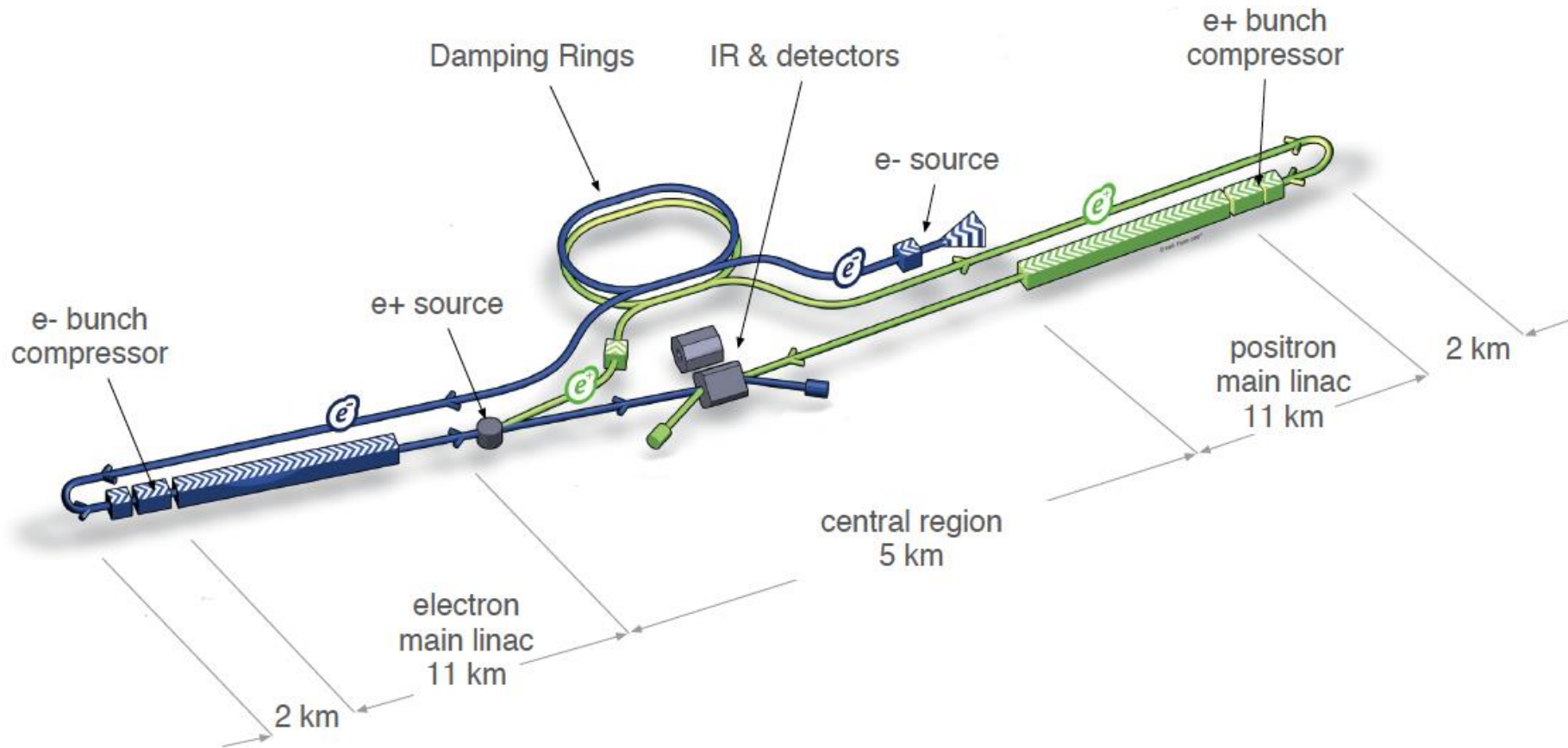


Low emittances are produced in the damping rings

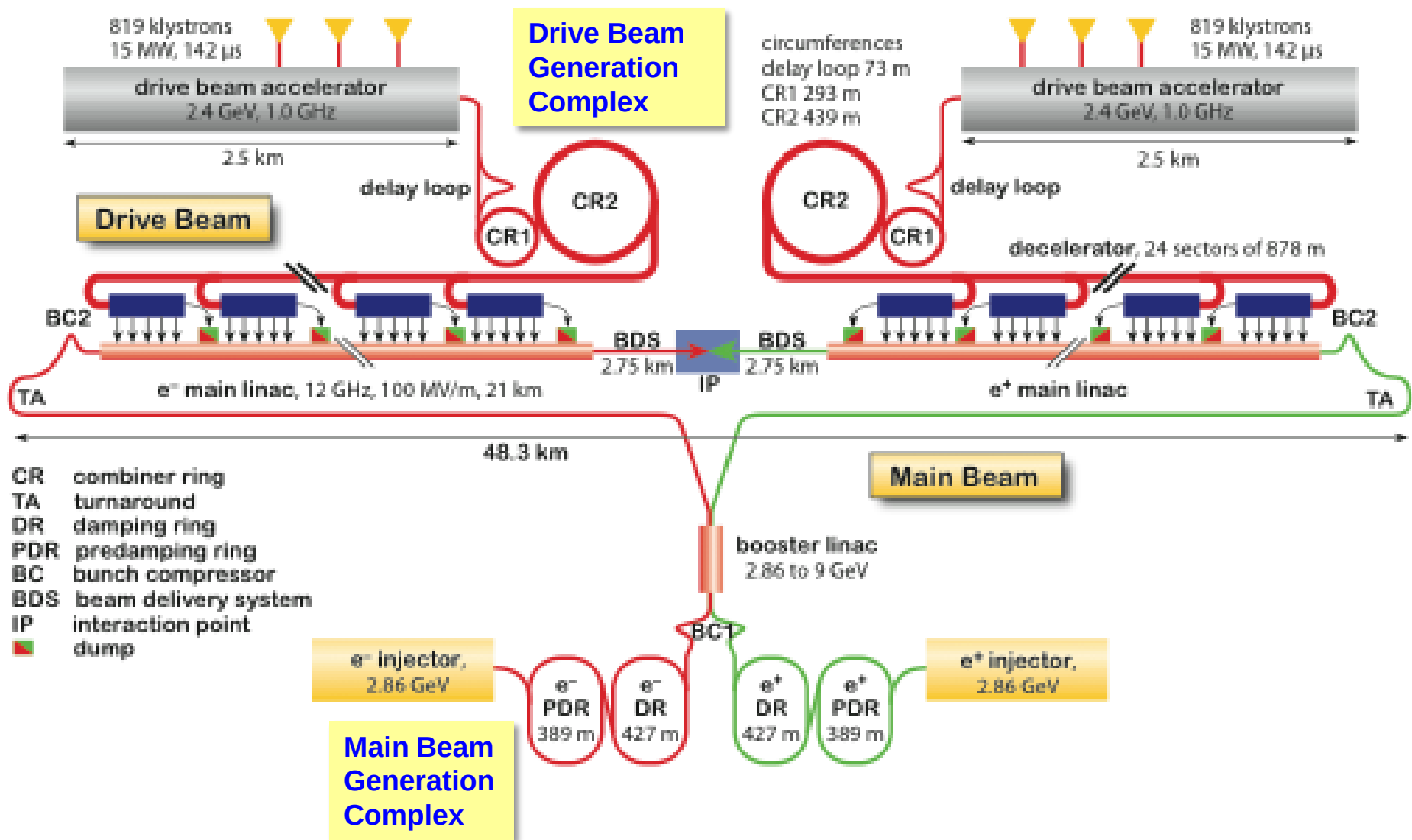
The must be maintained with limited degradation

The beam delivery system (BDS) squeezes the beam as much as possible, i.e. reduces $\beta_{x,y}$

ILC Layout



CLIC Layout at 3 TeV



Cavity/Accelerating Structure

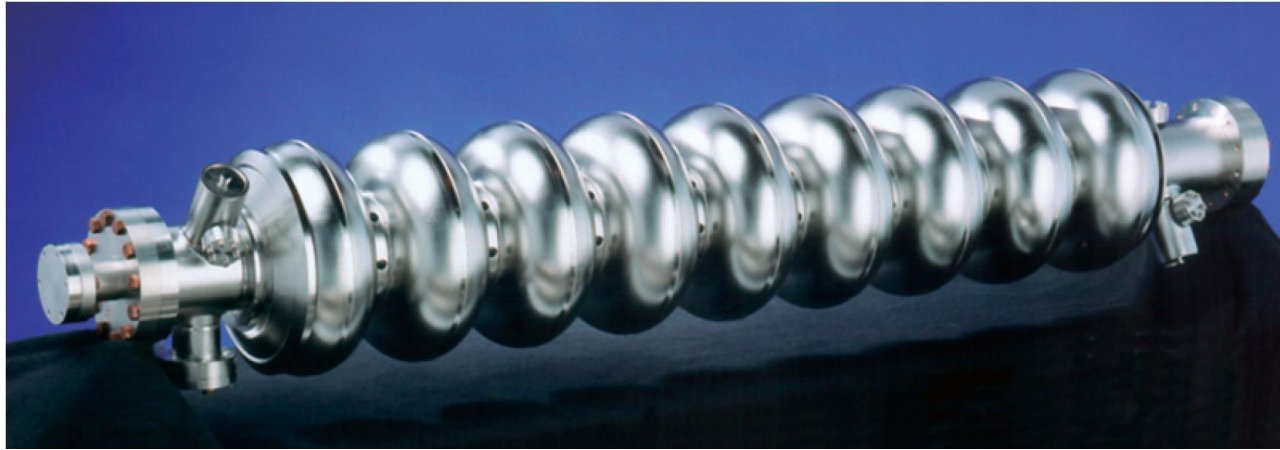
ILC cavity

1.3 GHz, superconducting

Target effective operational
31.5MV/m

Target gradient 35MV/m

$Q_0 \approx 10^{10}$



CLIC accelerating structure

12 GHz, normal conducting

Target loaded gradient 100MV/m

Target unloaded gradient 120MV/m

$Q_0 \approx 6 \cdot 10^3$

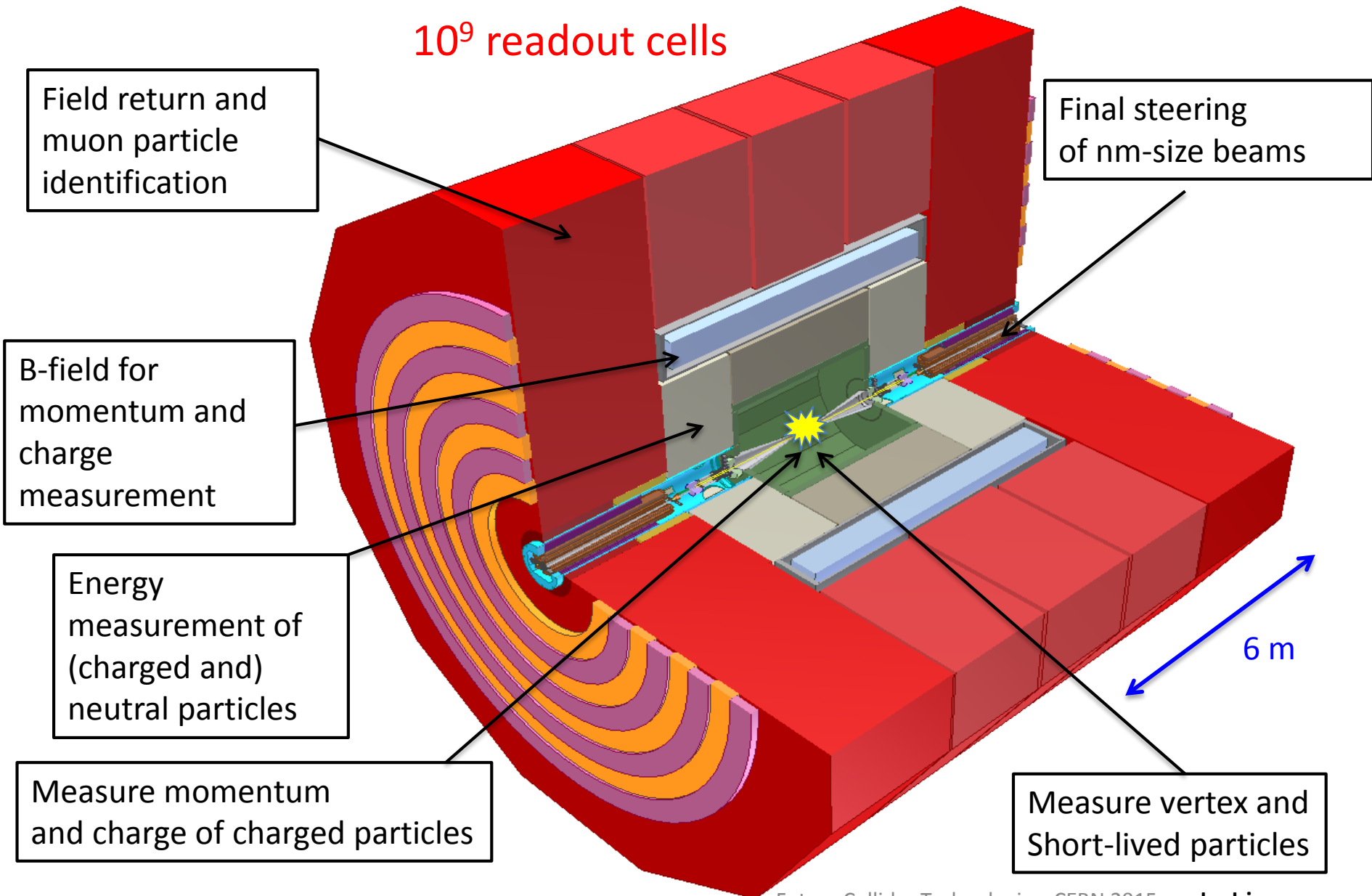
ILC and CLIC Main Parameters

Parameter	Symbol [unit]	SLC	ILC	CLIC	CLIC
Centre of mass energy	E_{cm} [GeV]	92	500	380	3000
Total luminosity	L [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	0.0003	1.8	1.5	6
Luminosity in peak	$L_{0.01}$ [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	0.0003	1	0.9	2
Gradient	G [MV/m]	20	31.5	72	100
Particles per bunch	N [10^9]	37	20	5.2	3.72
Bunch length	σ_z [μm]	1000	300	70	44
Collision beam size	$\sigma_{x,y}$ [nm/nm]	1700/600	474/5.9	143/2.9	40/1
Emittance	$\epsilon_{x,y}$ [$\mu\text{m}/\text{nm}$]	$\sim 3/3000$	10/35	0.95/30	0.66/20
Betafunction	$\beta_{x,y}$ [mm/mm]	$\sim 100/10$	11/0.48	8.2/0.1	6/0.07
Bunches per pulse	n_b	1	1312	352	312
Distance between bunches	Δz [mm]	-	554	0.5	0.5
Repetition rate	f_r [Hz]	120	5	50	50

There are more parameter sets for ILC and CLIC at different energies

CLIC at 3TeV has higher order optics and radiation effects

Linear Collider Experiment



Luminosity and Parameter Drivers

Can re-write normal
luminosity formula

$$\mathcal{L} = H_D \frac{N^2}{4\pi\sigma_x\sigma_y} n_b f_r$$

$$\mathcal{L} \propto H_D \underbrace{\frac{N}{\sigma_x}}_{\substack{\text{Luminosity} \\ \text{spectrum}}} \underbrace{N n_b f_r}_{\substack{\text{Beam power}}} \underbrace{\frac{1}{\sigma_y}}_{\substack{\text{Beam Quality} \\ (+\text{bunch length})}}$$

Somewhat simplified view

Beam-beam Effect

Bunches are squeezed strongly to maximise luminosity



Electron magnetic fields are very strong



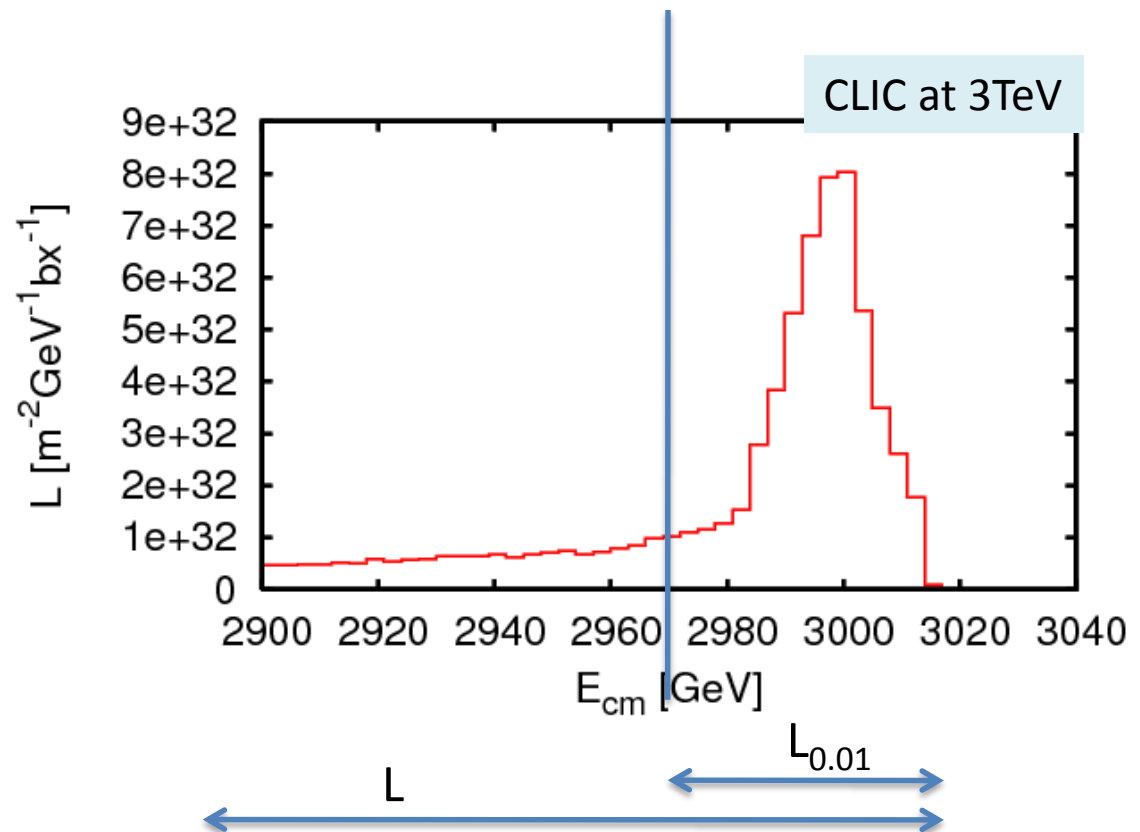
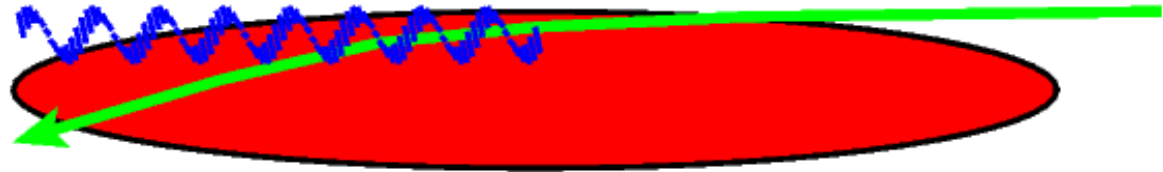
Beam particles travel on curved trajectories



They emit photons (O(1)) (beamstrahlung)



They collide with less than nominal energy



Beam Focusing

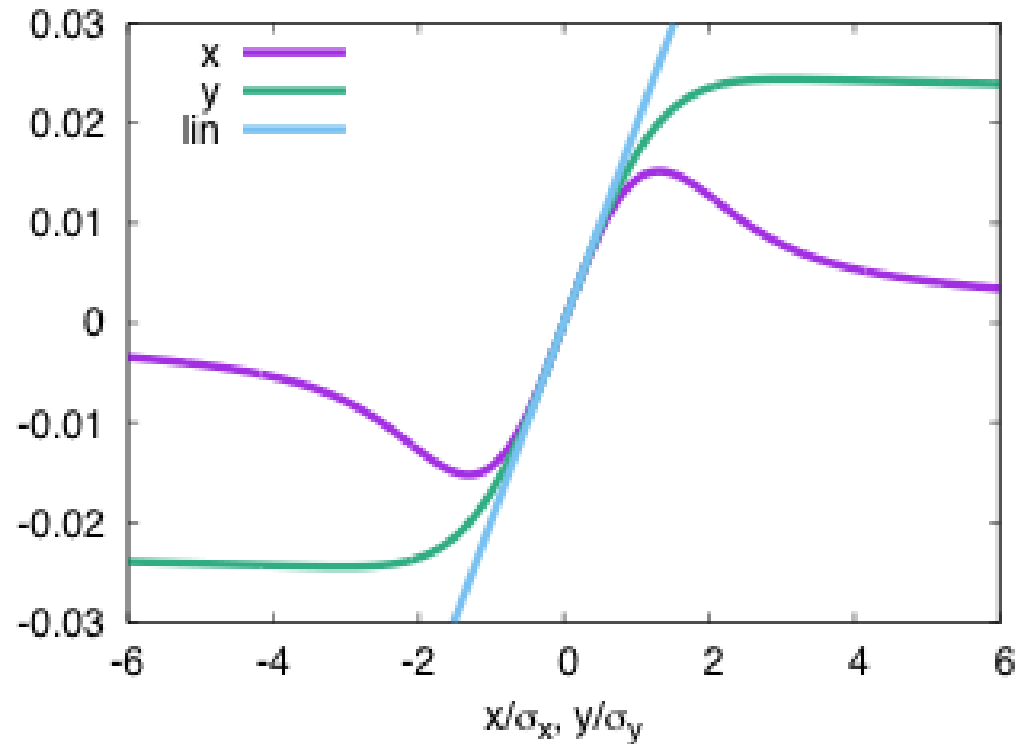
Deflection easy to calculate for

- small offset to the axis
- no initial angle
- negligible change of trajectory

Using previous beam-beam lecture

$$\left. \frac{dx}{dz} \right|_{final} = - \frac{2Nr_e x}{\gamma \sigma_x (\sigma_x + \sigma_y)}$$

$$\left. \frac{dy}{dz} \right|_{final} = - \frac{2Nr_e y}{\gamma \sigma_y (\sigma_x + \sigma_y)}$$



$$r_e \approx 2.8 \times 10^{-15} \text{ m}$$

Disruption Parameter

In each plane core of beam is focused to one point

$$s = f_x$$

$$s = f_y$$

We define the disruption parameter

$$D_x = \frac{\sigma_z}{f_x} = \frac{2Nr_e\sigma_z}{\gamma(\sigma_x + \sigma_y)\sigma_x}$$

$$D_y = \frac{\sigma_z}{f_y} = \frac{2Nr_e\sigma_z}{\gamma(\sigma_x + \sigma_y)\sigma_y}$$

$$D \ll 1$$

Particles do not move much in beam
⇒ Thin lens approximation
⇒ weak-strong simulation sufficient

$$D \gg 1$$

Particles do move in beam
⇒ Thick lens approximation
⇒ strong-strong simulation required

Typical Disruption

Parameter	Symbol [unit]	SLC	ILC	CLIC	CLIC
Centre of mass energy	E_{cm} [GeV]	92	500	380	3000
Particles per bunch	N [10^9]	37	20	5.2	3.72
Bunch length	σ_z [μm]	1000	300	70	44
Collision beam size	$\sigma_{x,y}$ [nm/nm]	1700/600	474/5.9	149/2.9	40/1
Vertical emittance	$\varepsilon_{x,y}$ [nm]	3000	35	40	20
Horizontal disruption	D_x	0.6	0.3	0.24	0.2
Vertical disruption	D_y	1.7	24.3	12.5	7.6

$$D_x \ll 1 \text{ and } D_y \gg 1$$



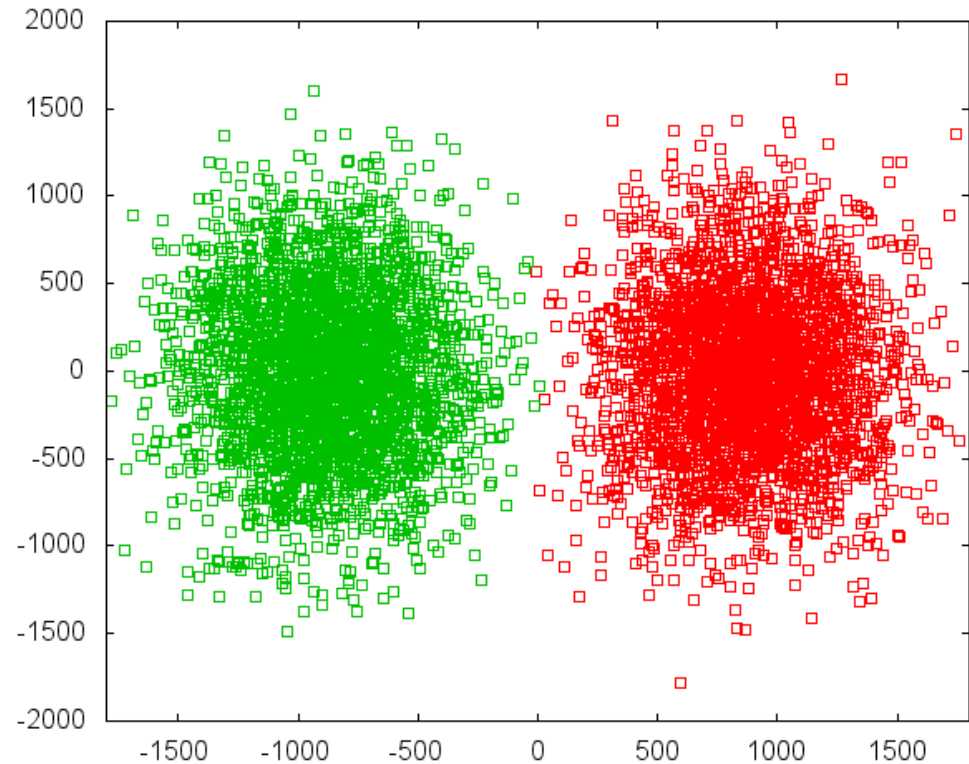
Need to resort to simulation

Simulation Codes

Need strong-strong code

- CAIN (K. Yokoya et al.)
- GUINEA-PIG (D. Schulte et al.)
- Beams => macro particles
- beams => slices
- Slices => cells
- The simulation is performed in a number of time steps in each of them
- The macro-particle charges are distributed over the cells
 - The forces at the cell locations are calculated
 - The forces are applied to the macro particles
 - The particles are advanced

X direction



Z direction

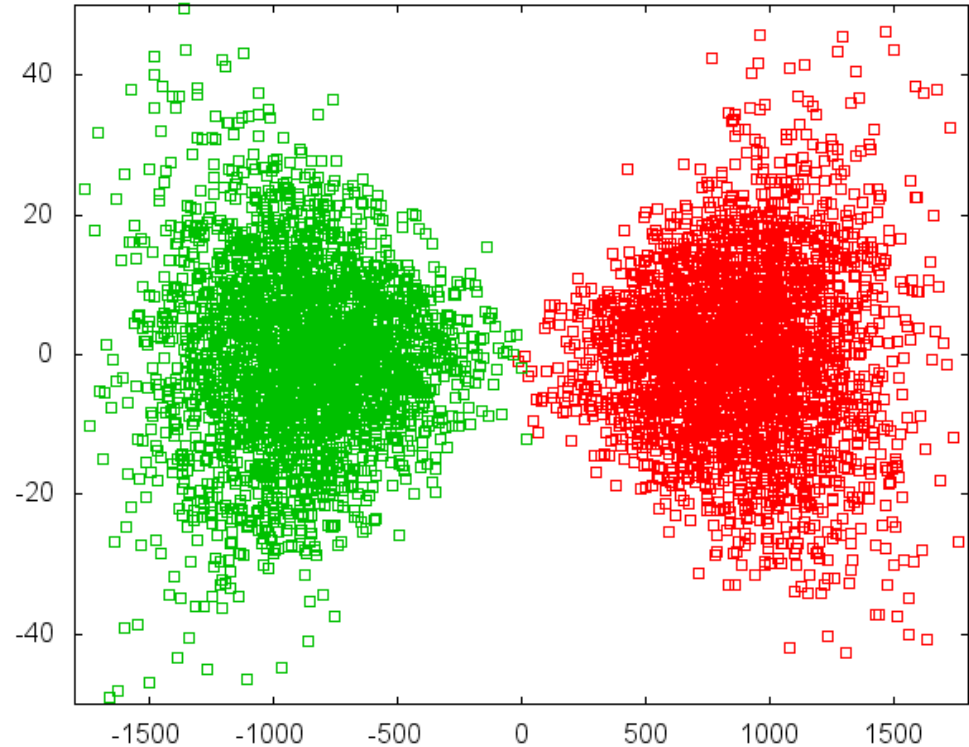
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X direction

Beam-beam force switched off



Z direction

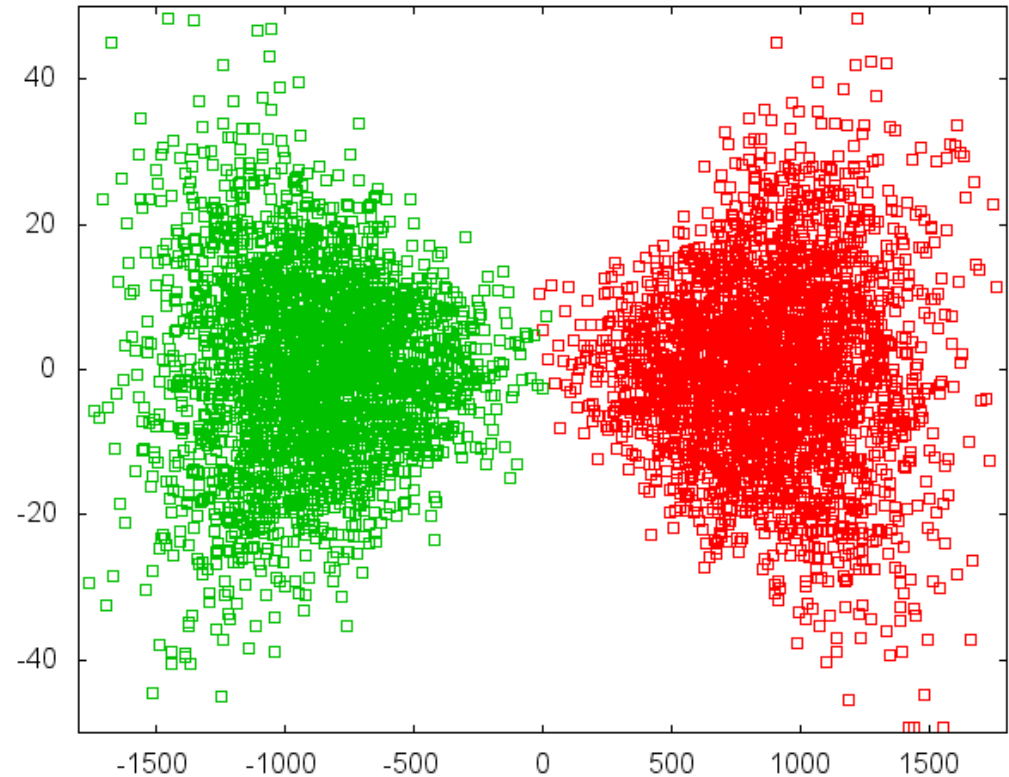
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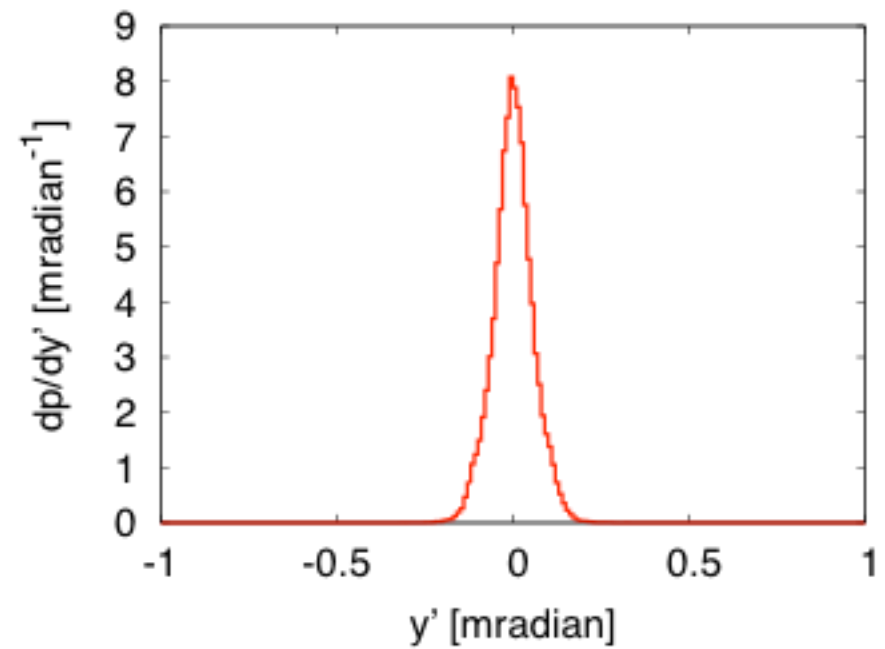
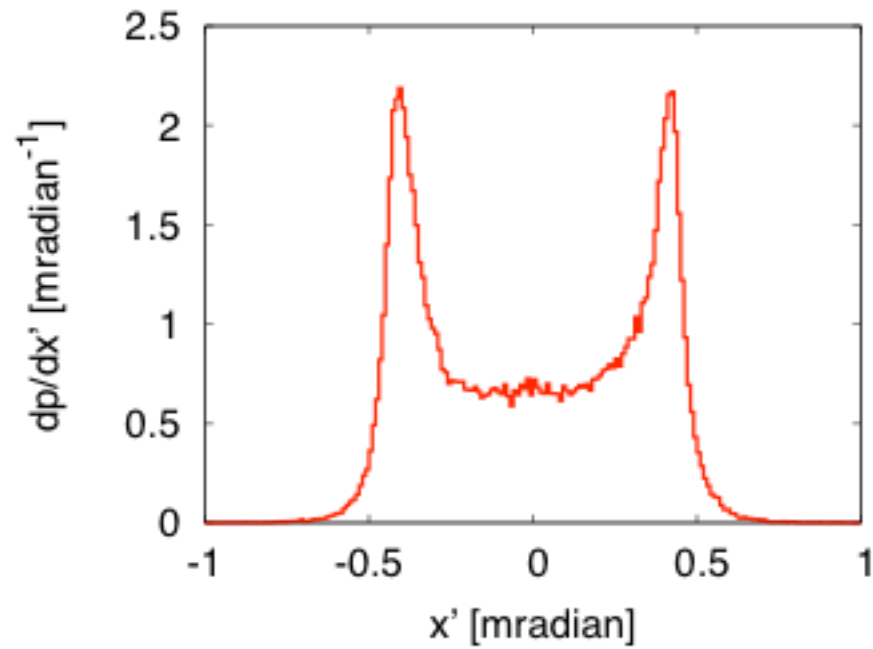
Y direction

Beam-beam force switched on



Z direction

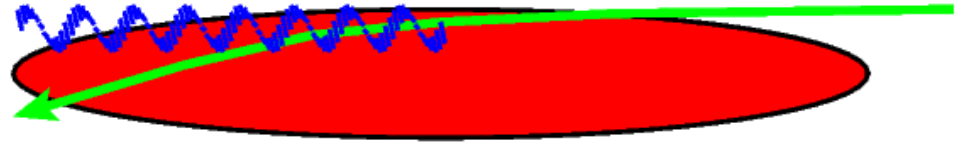
The Spent Beam



ILC and CLIC Main Parameters

Parameter	Symbol [unit]	ILC	CLIC	CLIC
Centre of mass energy	E_{cm} [GeV]	500	380	3000
Total luminosity	L [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	1.8	1.5	6
Luminosity in peak	$L_{0.01}$ [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	1	0.9	2
Particles per bunch	N [10^9]	20	5.2	3.72
Bunch length	σ_z [μm]	300	70	44
Collision beam size	$\sigma_{x,y}$ [nm/nm]	474/5.9	149/2.9	40/1
Vertical emittance	$\epsilon_{x,y}$ [nm]	35	40	20
Geometric luminosity	L_{geom} [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	0.75	0.8	4.3
Enhancement factor	H_D	2.4	1.9	1.5

Beamstrahlung



Similar to synchrotron radiation

From local trajectory curvature calculate critical

$$\hbar\omega_c = \frac{3}{2} \frac{\gamma^3 \hbar c}{\rho}$$

Define beamstrahlung parameter Upsilon

$$\Upsilon = \frac{2}{3} \frac{\hbar\omega_c}{E_0}$$

Average Upsilon is approximately given by

$$\langle \Upsilon \rangle = \frac{5}{6} \frac{N r_e^2 \gamma}{\alpha (\sigma_x + \sigma_y) \sigma_z}$$

Beamstrahlung Spectrum

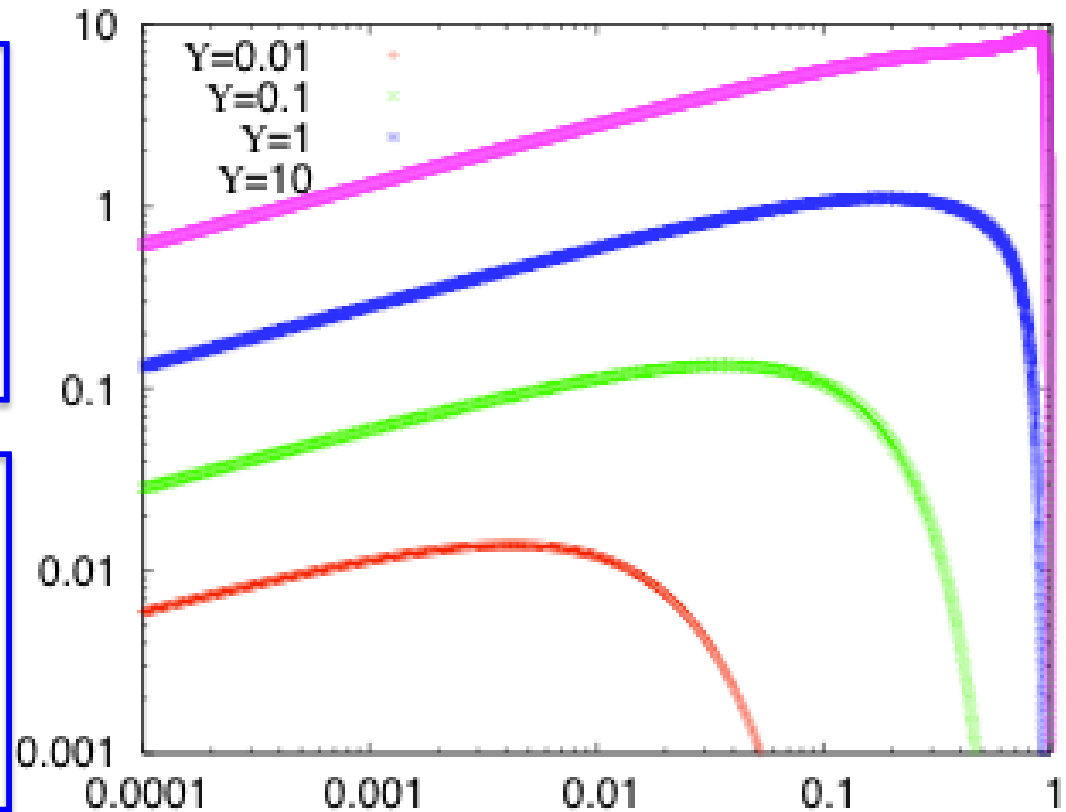
Classical regime $\Upsilon \ll 1$

ILC 0.06

CLIC at 380GeV 0.17

Quantum regime $\Upsilon \gg 1$

CLIC at 3TeV 5



The spectrum is given by
$$\frac{d\dot{w}}{d\omega} = \frac{\alpha}{\sqrt{3}\pi\gamma^2} \left[\int_x^\infty K_{\frac{5}{3}}(x') dx' + \frac{\hbar\omega}{E} \frac{\hbar\omega}{E - \hbar\omega} K_{\frac{2}{3}}(x) \right]$$

With modified Bessel functions

$K_{\frac{5}{3}}$ and

$K_{\frac{2}{3}}$

$$x = \frac{\omega}{\omega_c} \frac{E}{E - \hbar\omega}$$

Photons in the Classical Regime

$$\Upsilon \ll 1$$

Number of photons
Dominates $L/L_{0.01}$

$$n_\gamma \propto \frac{N}{\sigma_x + \sigma_y}$$

$$\mathcal{L} \propto \frac{N}{\sigma_x \sigma_y}$$

Energy of photons
Defines shape of tail

$$\frac{E_\gamma}{E_0} \propto \frac{N}{\sigma_x + \sigma_y} \frac{\gamma}{\sigma_z}$$

So use

$$\sigma_x \gg \sigma_y$$

$$\sigma_x + \sigma_y \approx \sigma_x$$

$$\mathcal{L} \propto H_D \left(\frac{N}{\sigma_x} \right) N n_b f_r \frac{1}{\sigma_y}$$

Determined by beamstrahlung

ILC and CLIC Main Parameters

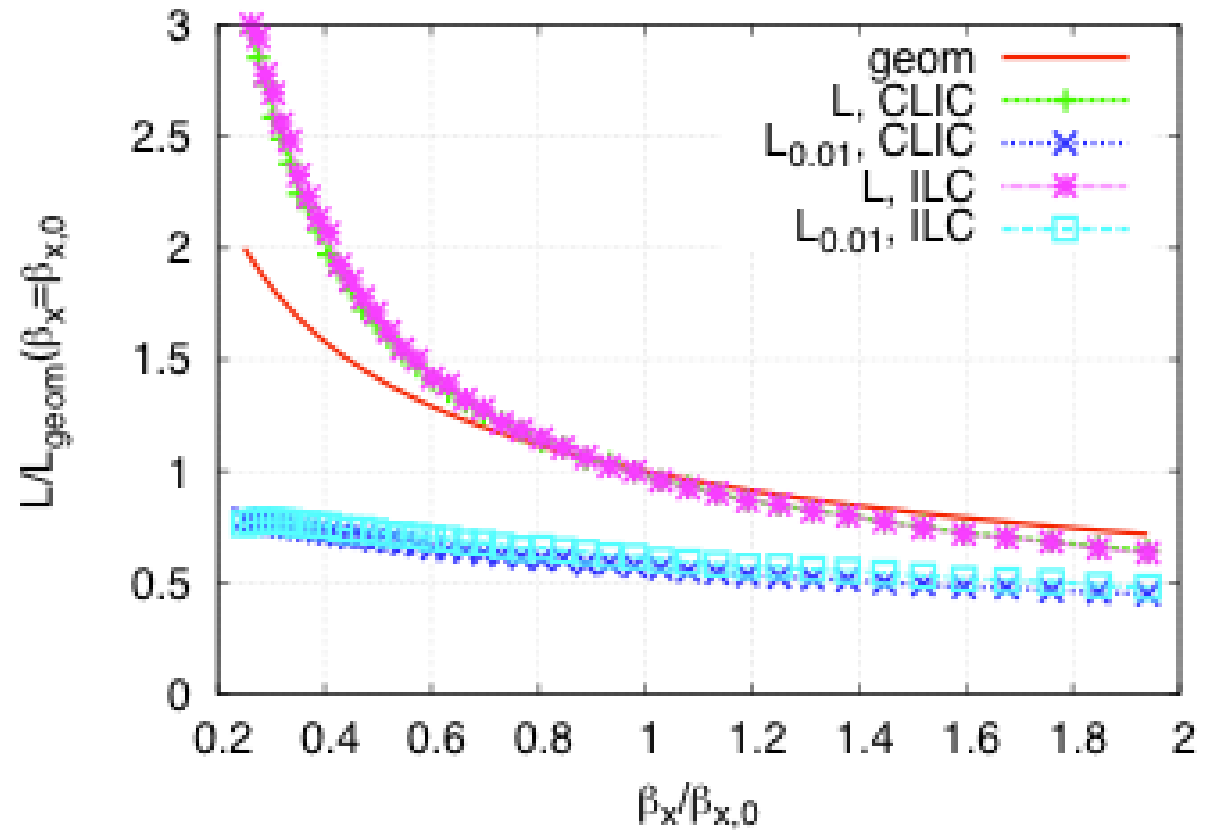
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Total luminosity	L [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	1.8	1.5	6
Luminosity in peak	$L_{0.01}$ [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	1	0.9	2
Particles per bunch	N [10^9]	20	5.2	3.72
Bunch length	σ_z [μm]	300	70	44
Collision beam size	$\sigma_{x,y}$ [nm/nm]	474/5.9	149/2.9	40/1
Vertical emittance	$\epsilon_{x,y}$ [nm]	35	40	20
Photons per beam particle	n_γ	1.9	1.5	2.1
Average photon energy	$\langle E_\gamma/E_0 \rangle$ [%]	2.4	4.5	13

Photon numbers and $L_{0.01}/L$ are similar for ILC and CLIC at low energies

Average photon energy does not seem to matter too much for $L_{0.01}$

Luminosity Spectrum

Vary horizontal beta-function
around nominal value



Reducing the horizontal beam
size pushes total luminosity

But $L_{0.01}$ does not increase
much

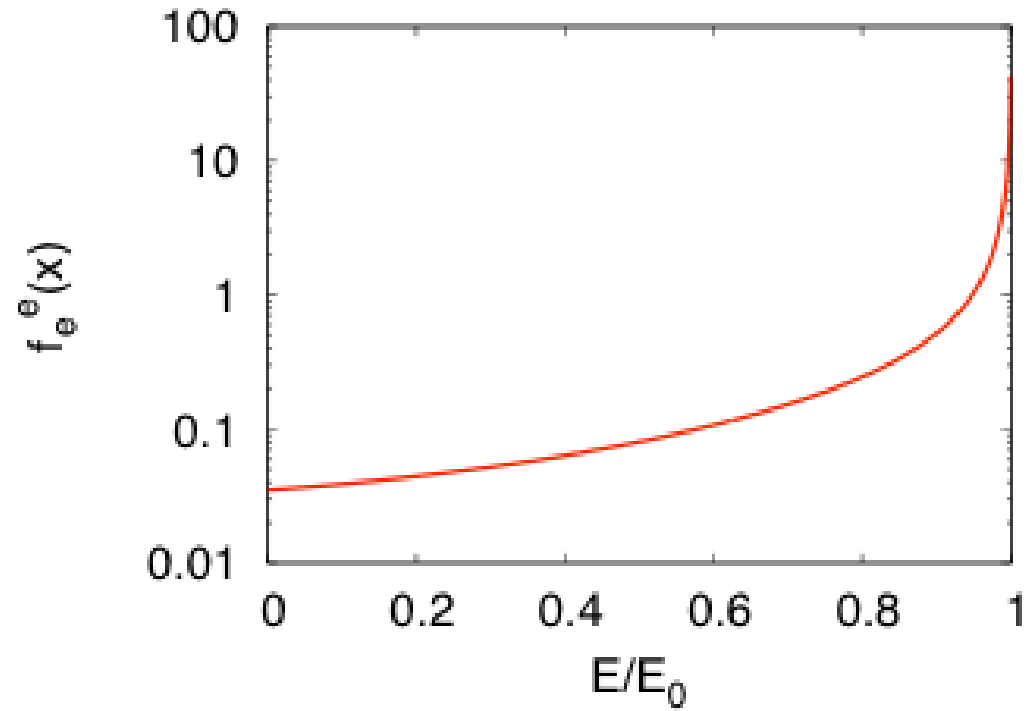
Initial State Radiation

Electrons often emit a photon before the collision

⇒ Initial State Radiation

The electron can be replaced by a spectrum of electrons

This yields a luminosity spectrum



If you really want some formulae:
This is what GUINEA-PIG uses

$$f_e^e(x, Q^2) = \frac{\beta}{2}(1-x)^{(\frac{\beta}{2}-1)} \left(1 + \frac{3}{8}\beta\right) - \frac{\beta}{4}(1+x)$$
$$\beta = \frac{2\alpha}{\pi} \left(\ln \frac{Q^2}{m^2} - 1 \right)$$

$$Q^2 = s = 4E_{cm}^2$$

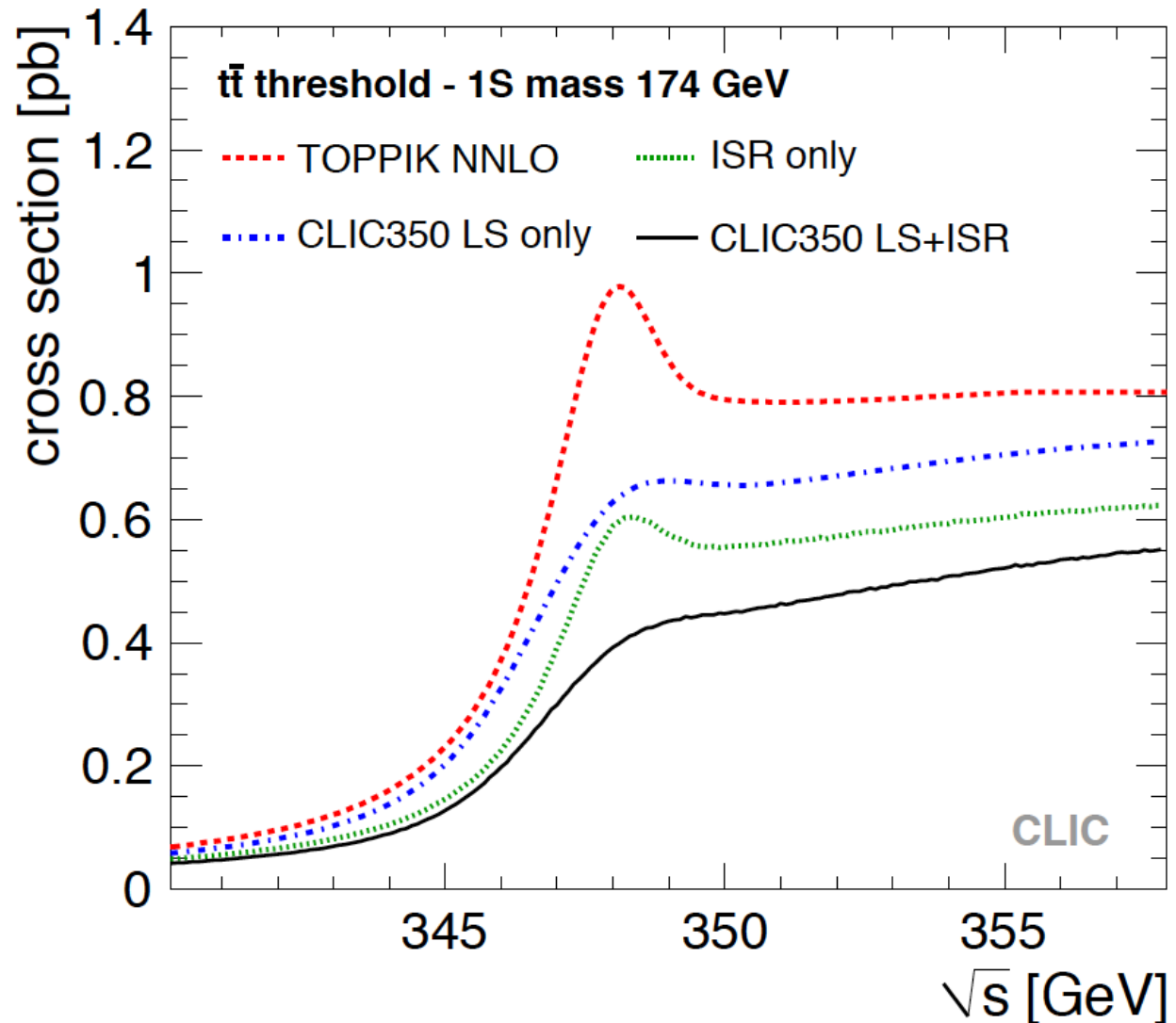
Top Production at Threshold

K. Seidel et al. [arXiv:1303.3758](#)

Top production at threshold
is strongly affected by beam
energy spread and
beamstrahlung

Typical requirement:
 $L_{0.01} > 0.6 L$

But depends on physics



Vertical Beamsize

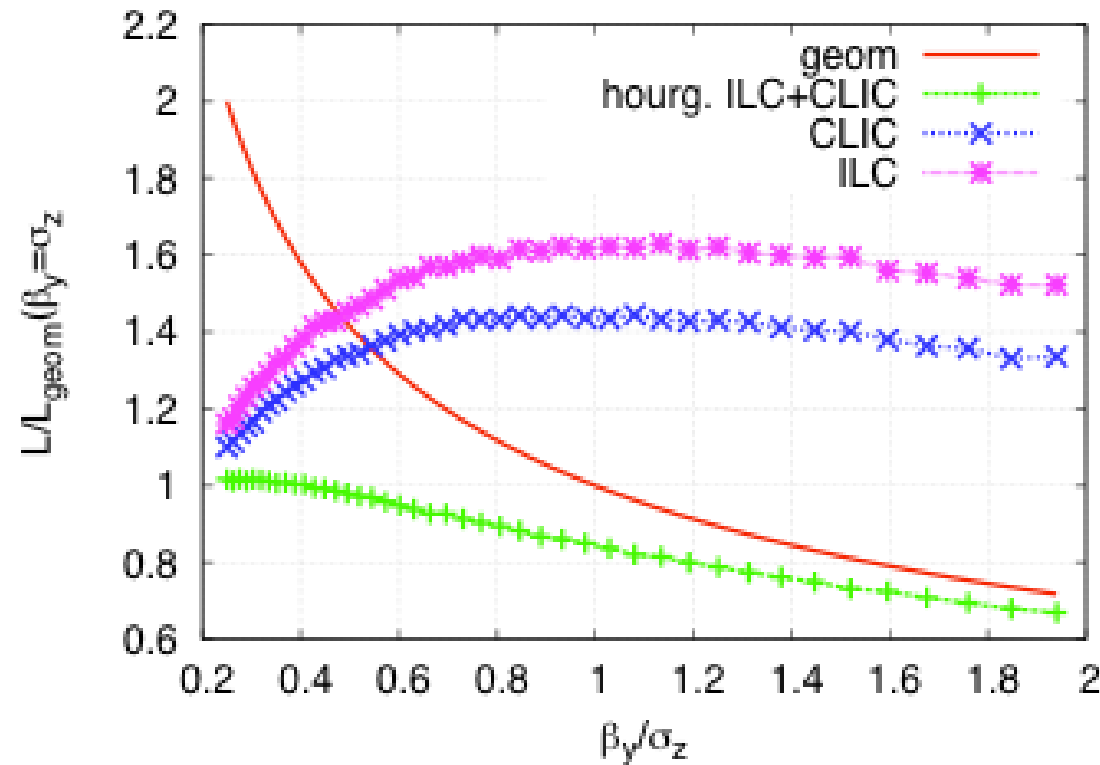
Beam size is given by

$$\mathcal{L} = \frac{N^2}{4\pi\sigma_x\sigma_y} n_b f_r$$

$$\sigma(s) = \sqrt{\frac{\beta(s)\epsilon}{\gamma}} \quad \beta(s) = \sqrt{\beta(0) + \frac{s^2}{\beta(0)}}$$

$$\mathcal{L} = H_D \frac{N^2}{4\pi\sigma_x\sigma_y} n_b f_r$$

Chose $\beta \geq \sigma_z$ Difficult to make



Note: Waist Shift

Quantum Regime

Beamstrahlung is suppressed at high energy

$$\Upsilon \gg 1$$

Number of photons is

$$n_\gamma \propto \Upsilon^{2/3} \frac{\sigma_z}{\gamma} \propto \left(\frac{N}{\sigma_x + \sigma_y} \right)^{2/3} \left(\frac{\sigma_z}{\gamma} \right)^{1/3}$$

Energy of photons is

$$\langle E_\gamma \rangle \approx 0.254 E_0$$

Useful approximate formulae:

Beamstrahlung Optimisation

For low energies (classical regime) number of emitted photons

$$n_\gamma \propto E_\gamma \propto \frac{N}{\sigma_x + \sigma_y}$$

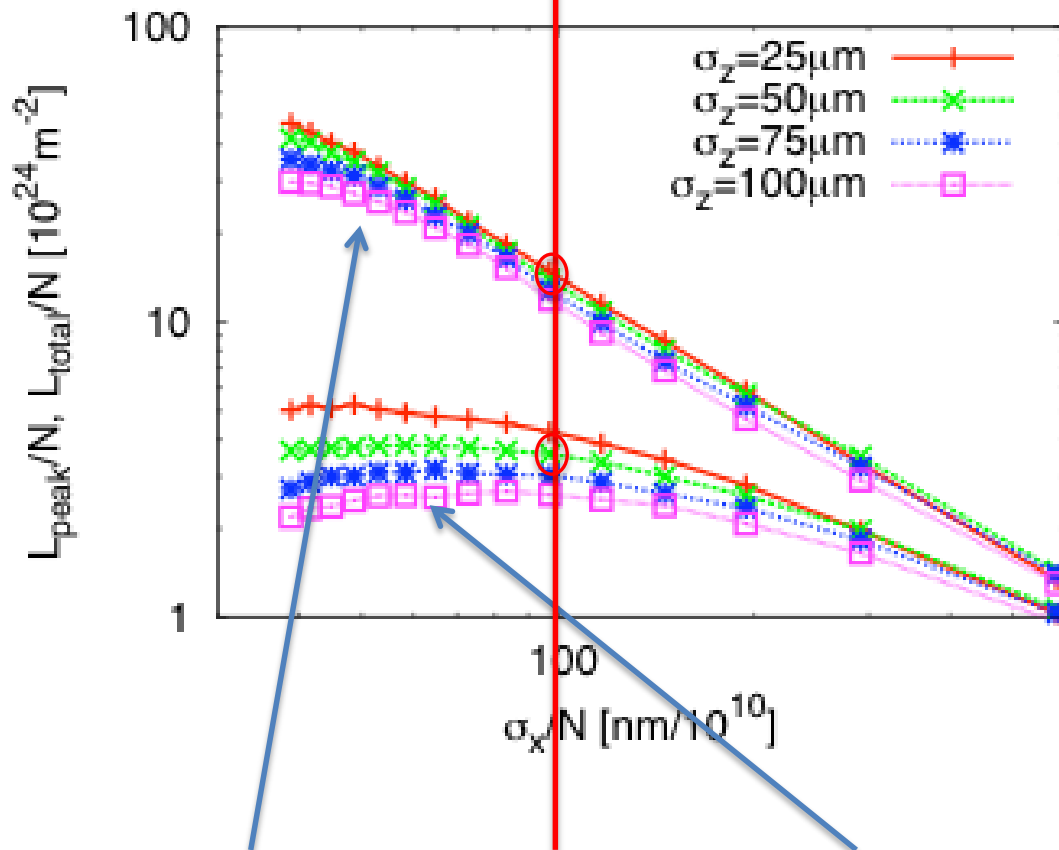
$$\mathcal{L} \propto \frac{N}{\sigma_x \sigma_y}$$

Hence use $\sigma_x \gg \sigma_y$

$$\sigma_x + \sigma_y \approx \sigma_x$$

$$\mathcal{L} \propto H_D \left(\frac{N}{\sigma_x} \right) N n_b f_r \frac{1}{\sigma_y}$$

For CLIC at 3 TeV (quantum regime)

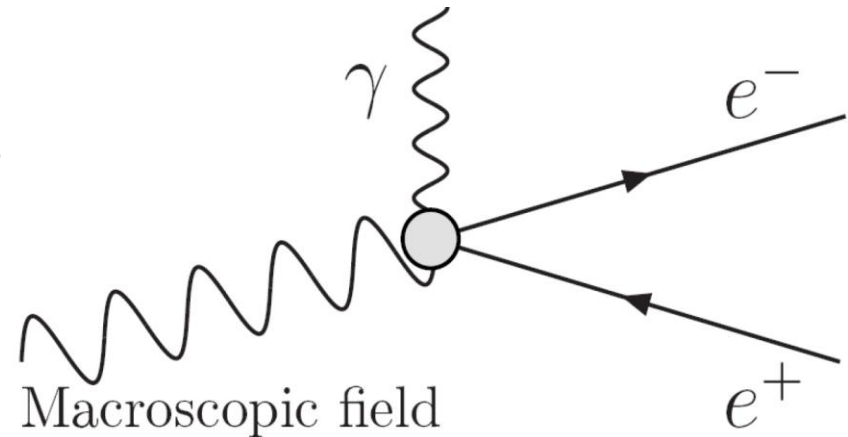


Total luminosity grows for smaller beams
CLIC parameter choice
luminosity in peak starts to decrease again

Coherent Pair Creation

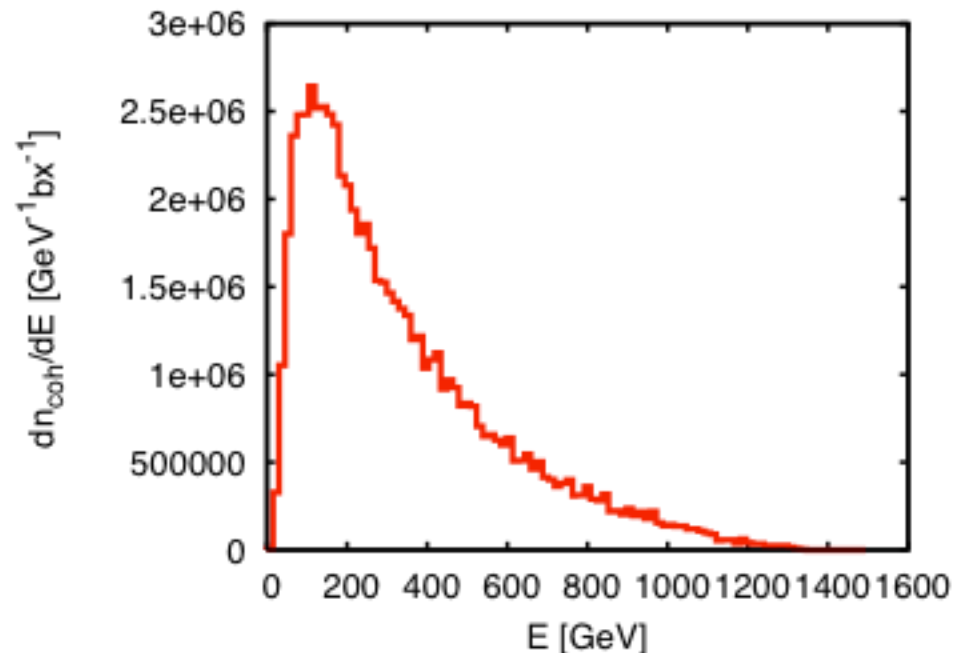
Beam fields in the rest system of a photon can reach the **Schwinger Critical Field**
⇒ The quantum electrodynamics becomes non-linear

A photon in a very strong field can form an electron-positron pair
⇒ Coherent pair creation



$$\frac{\gamma B}{B_c} = \Upsilon$$

$$B_c \approx 4.4 \times 10^9 \text{T}$$



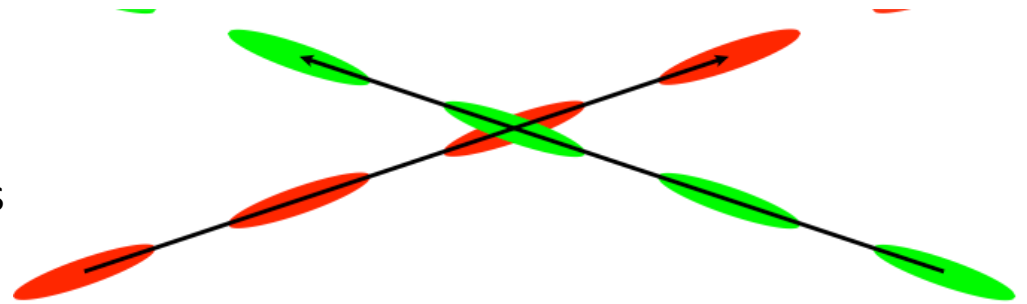
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Photons per beam particle	n_γ	1.9	1.5	2.1
Average photon energy	$\langle E_\gamma/E_0 \rangle$ [%]	2.4	4.5	13
Coherent pairs	N_{coh}	-	-	6.8×10^8
Their energy	E_{coh} [TeV]	-	-	2.1×10^8

Crossing Angle

Have crossing angles

- ILC: 14mradian
- CLIC: 20mradian
- to reduce effects of parasitic crossings
- to extract the spent beam cleanly

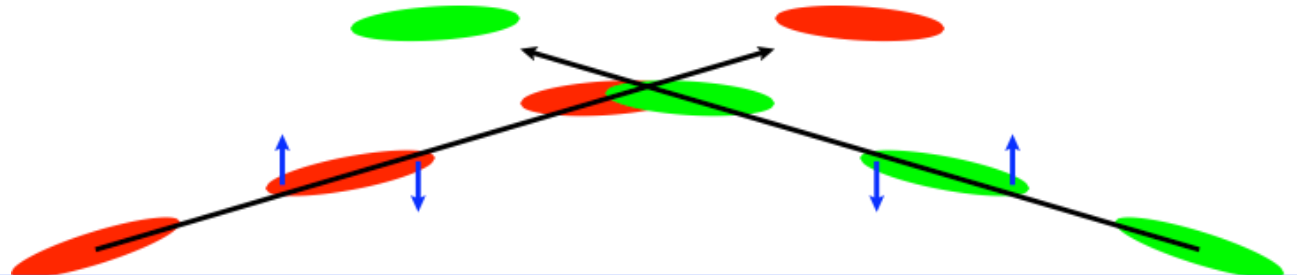


Luminosity with crossing angle

$$\mathcal{L} = H_D \frac{N^2 f_r n_b}{4\pi \sigma_x \sigma_y} \frac{1}{\sqrt{1 + \left(\frac{\sigma_z}{\sigma_x} \tan \frac{\theta_c}{2} \right)^2}}$$

0.1-0.2

Use crab cavities:



Can ignore crossing angle for beam-beam calculation
But not in detector design

Spent Beam Divergence

Beam particles are focused by oncoming beam

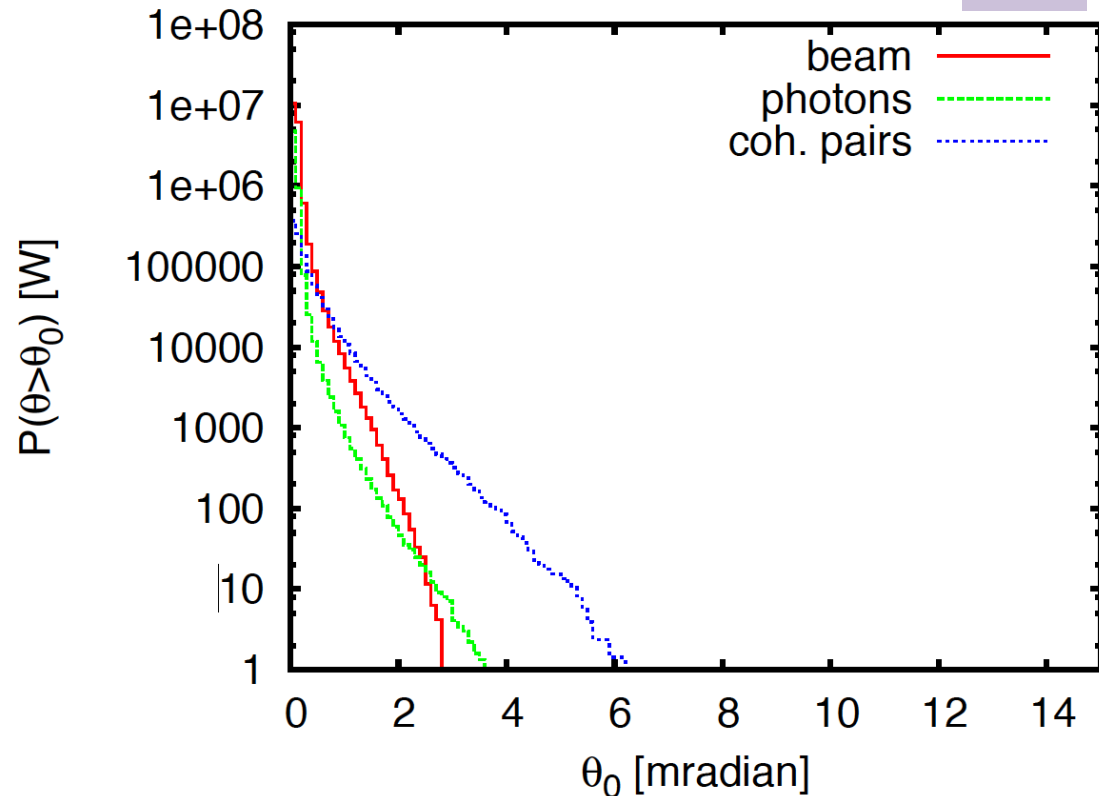
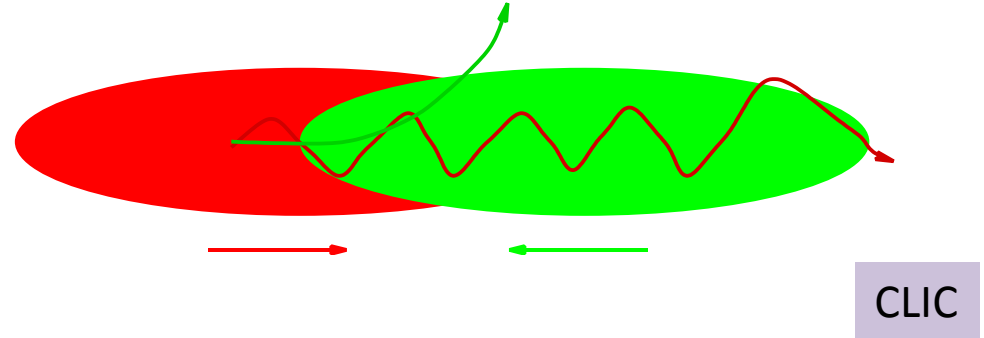
Photons are radiated into direction of beam particles

Coherent pair particles can be focused or defocused by the beams but deflection limited due to their high energy

-> Extraction hole angle should be significantly larger than 6mradian

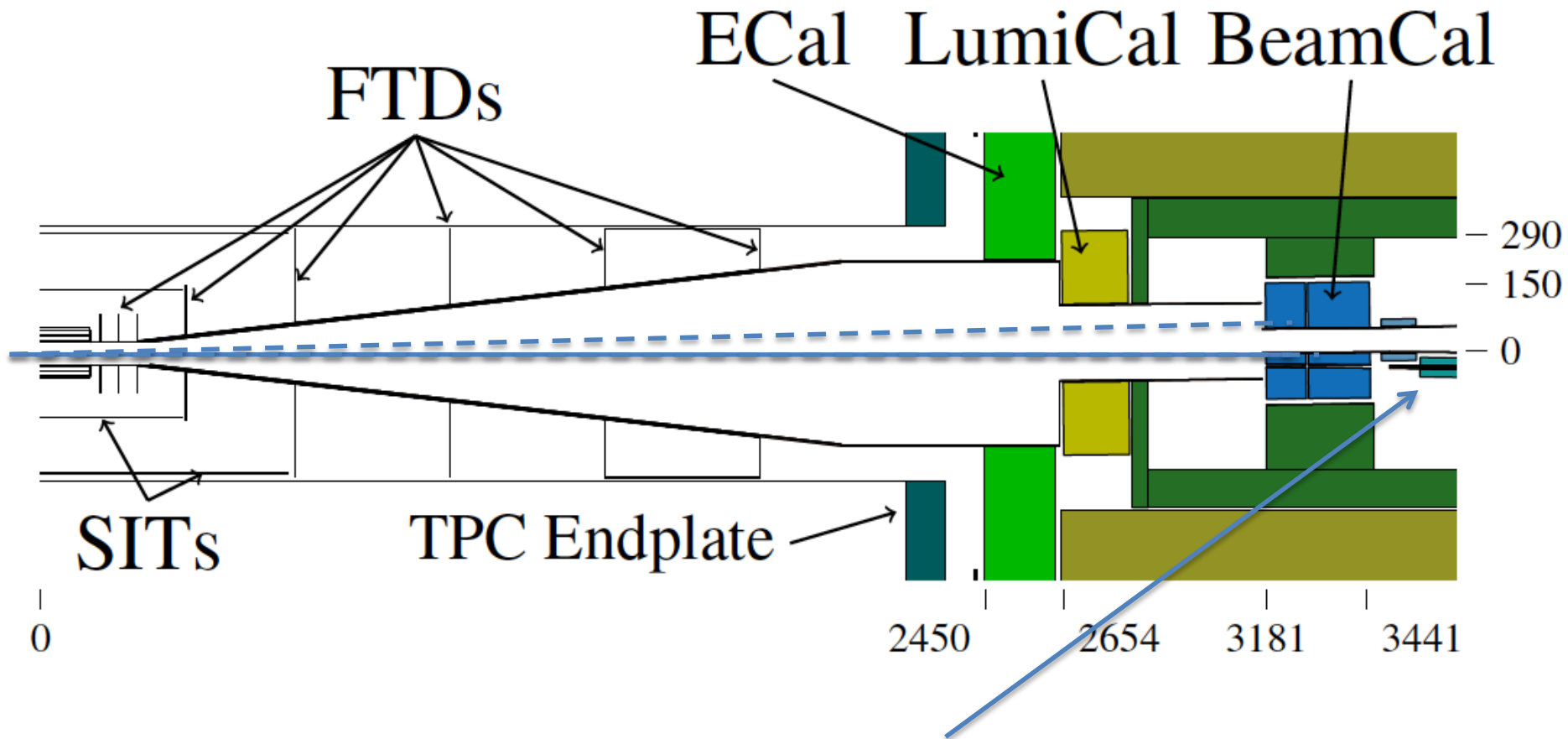
We chose 10mradian for CLIC
-> 20mradian crossing angle

ILC requires 14mradian crossing angle



1 W $\approx$ 400 TeV/bx $\approx$
300 beamparticles/bx

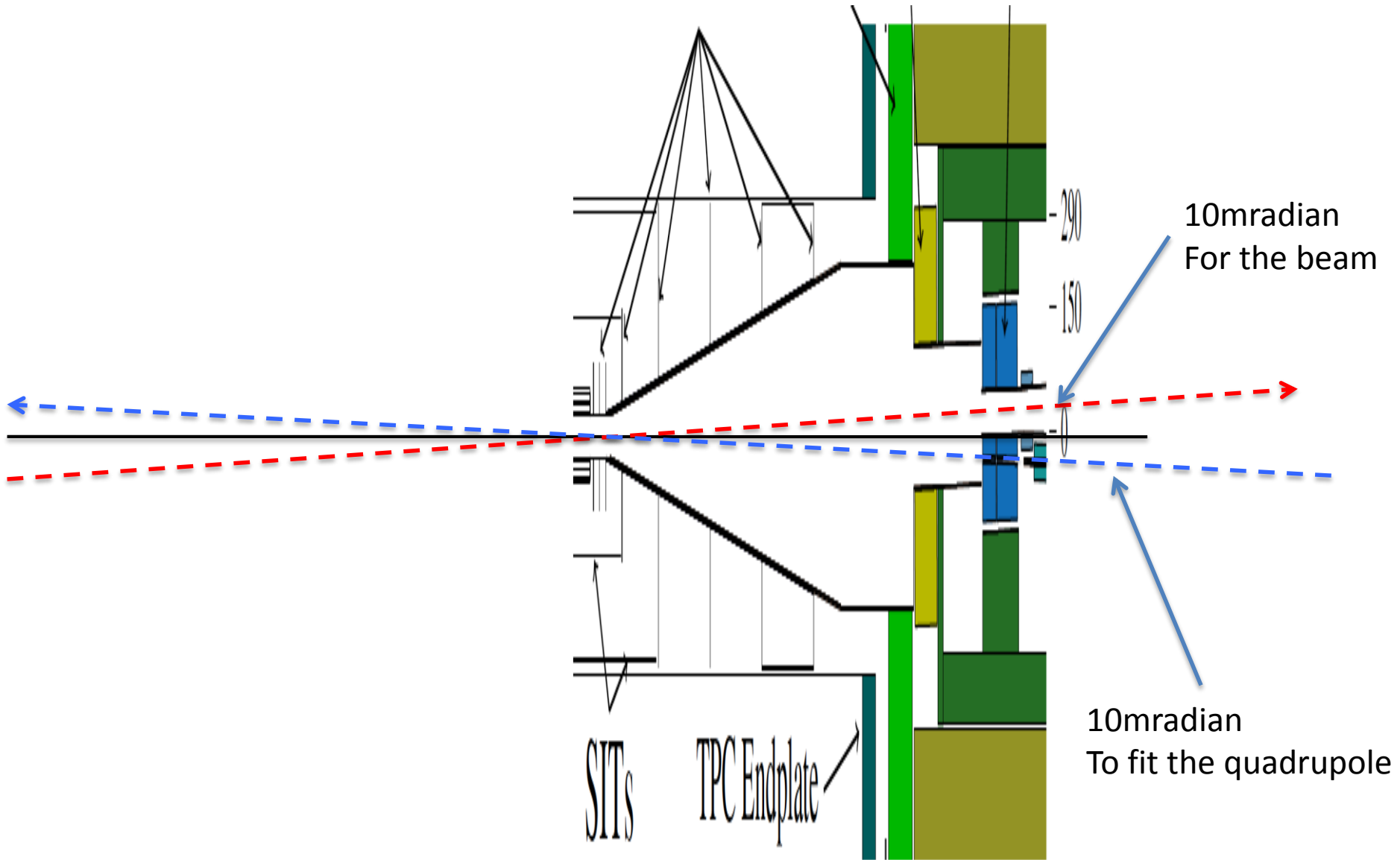
CLIC Inner Detector Layout



The last focusing magnet of the machine is inside of the detector

A. Seiler

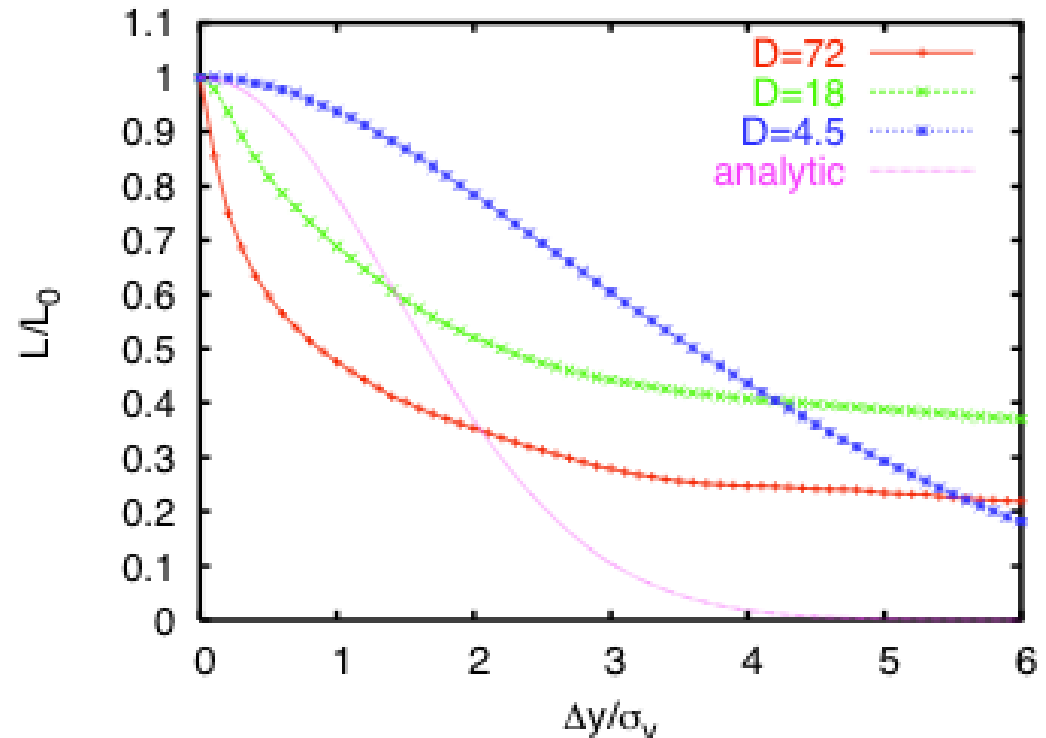
Inner Detector Layout



Luminosity and Offset

Luminosity loss for beam offsets depends strongly on disruption parameter

Can become a problem



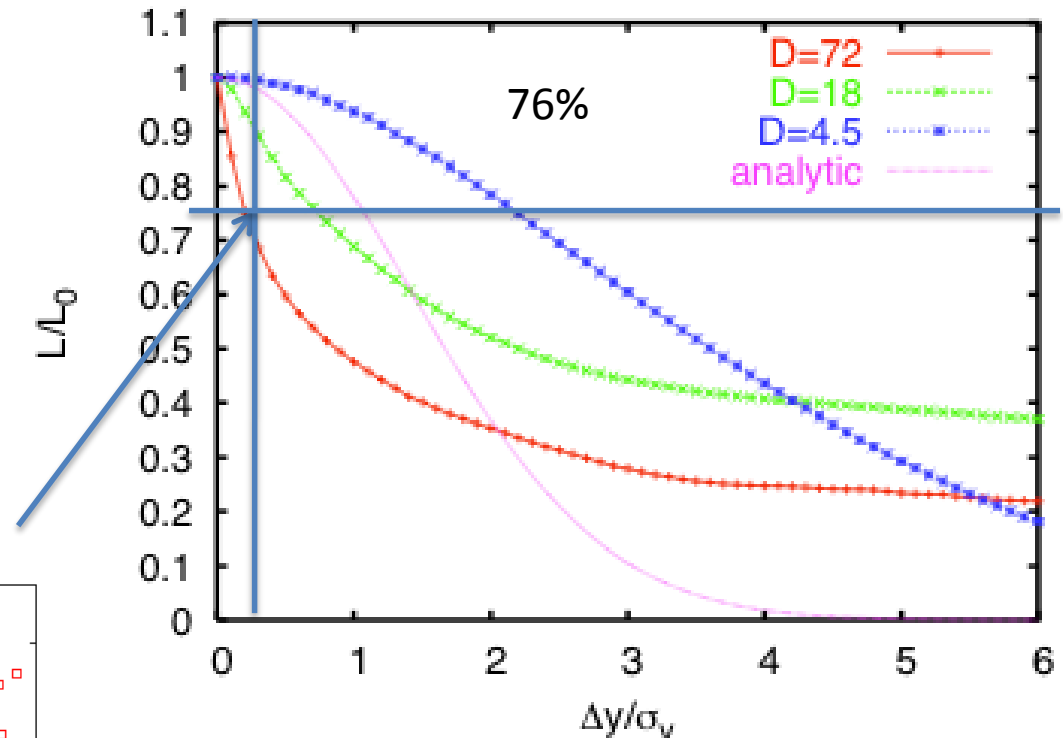
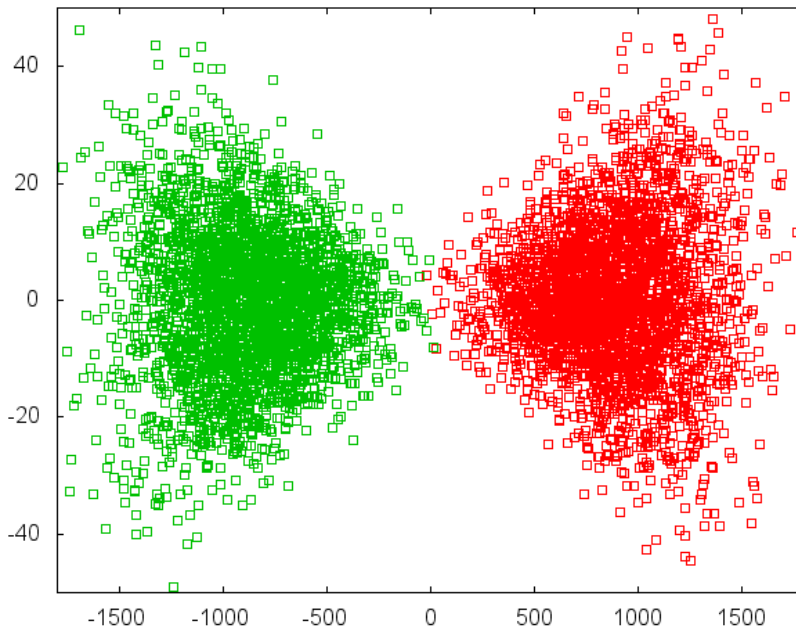
$$\frac{\mathcal{L}}{\mathcal{L}_0} = \exp\left(-\frac{\Delta y^2}{4\sigma_y^2}\right)$$

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ILC



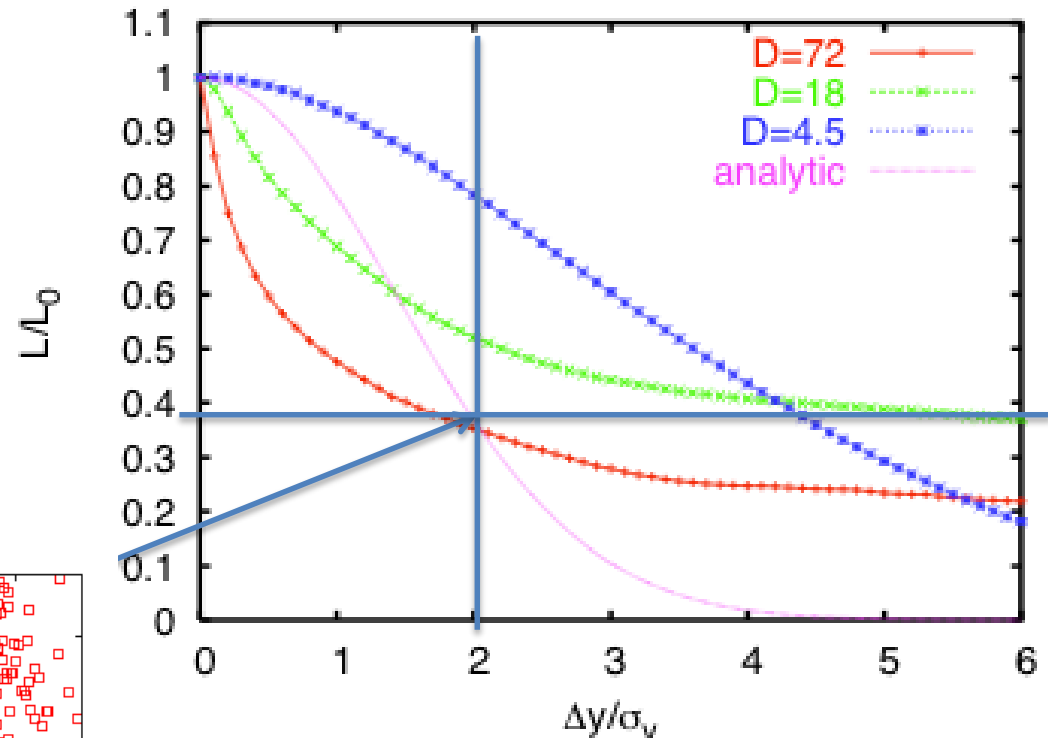
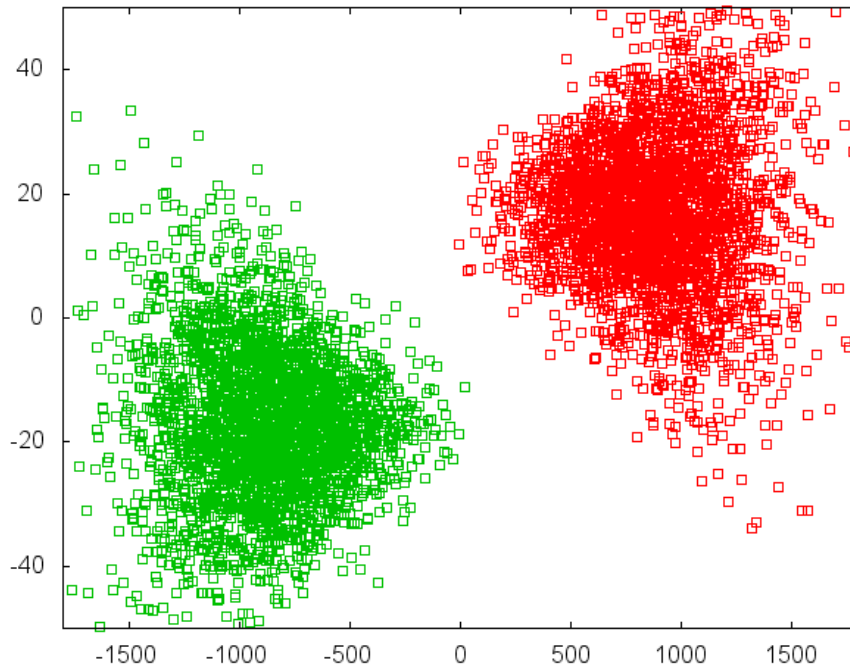
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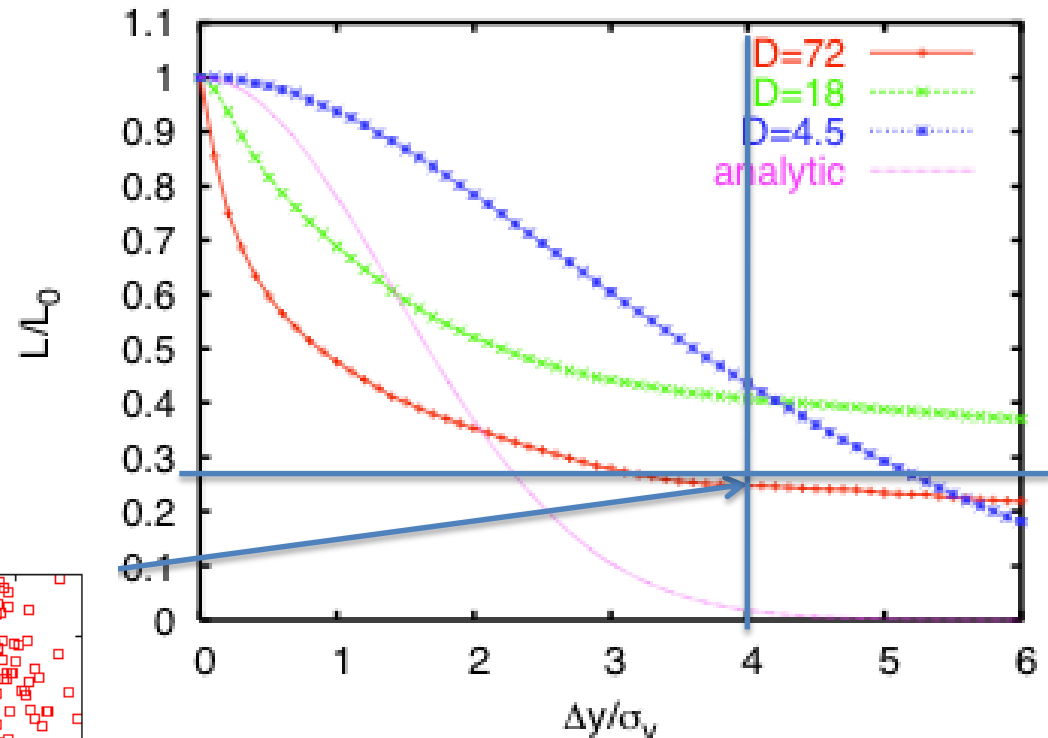
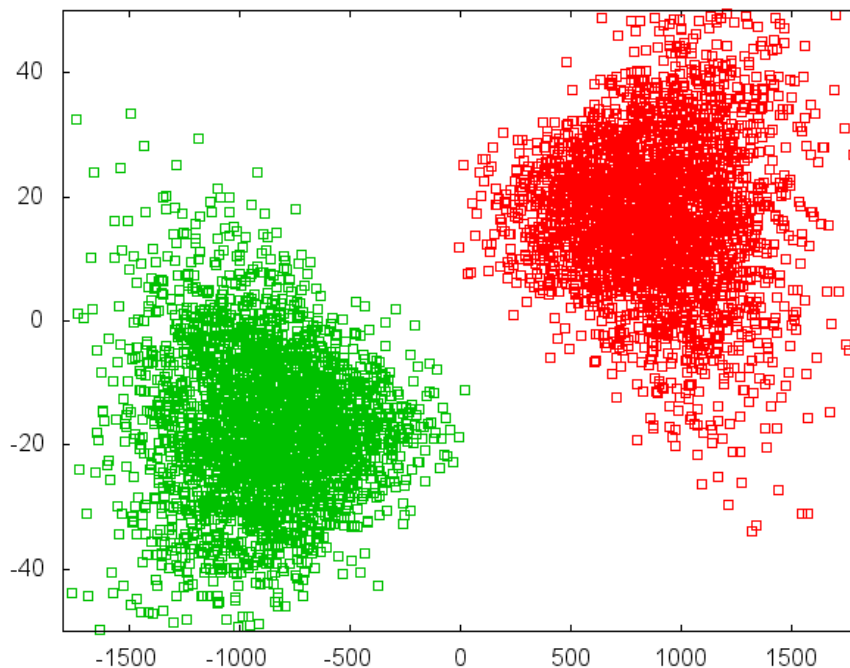
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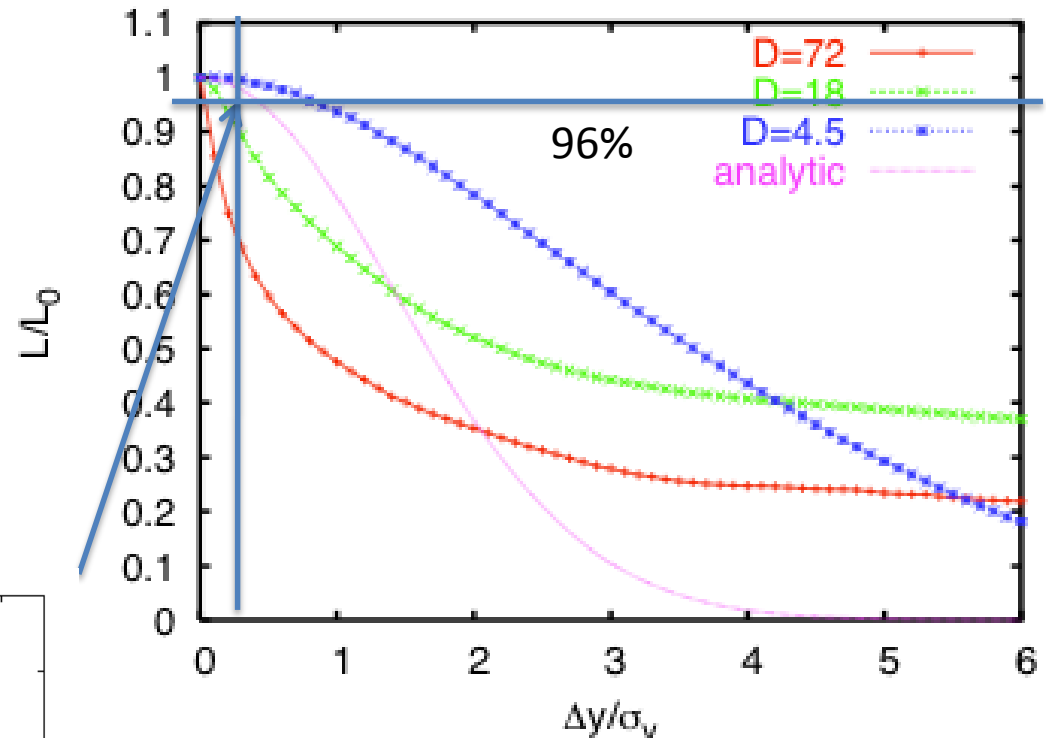
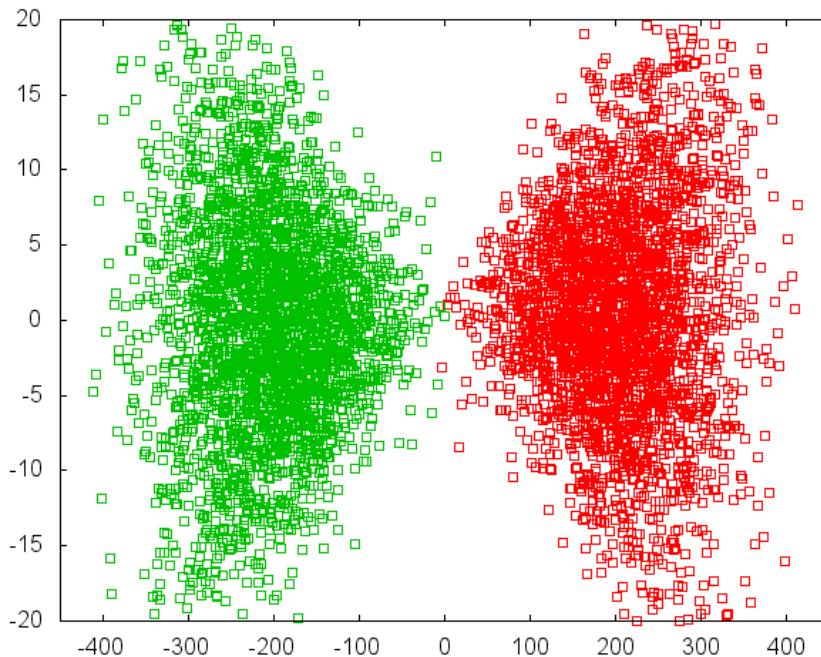
$$\frac{\mathcal{L}}{\mathcal{L}_0} = \exp\left(-\frac{\Delta y^2}{4\sigma_y^2}\right)$$

Luminosity and Offset

Luminosity loss for beam offsets depends strongly on disruption parameter

Can become a problem

CLIC at 380GeV



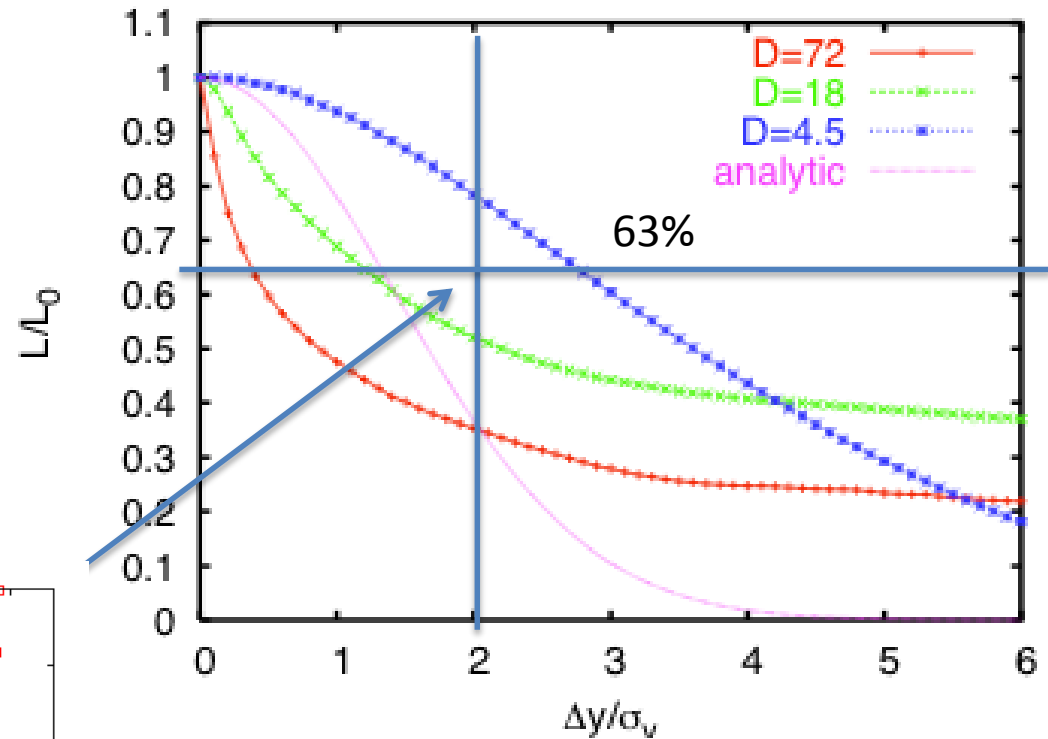
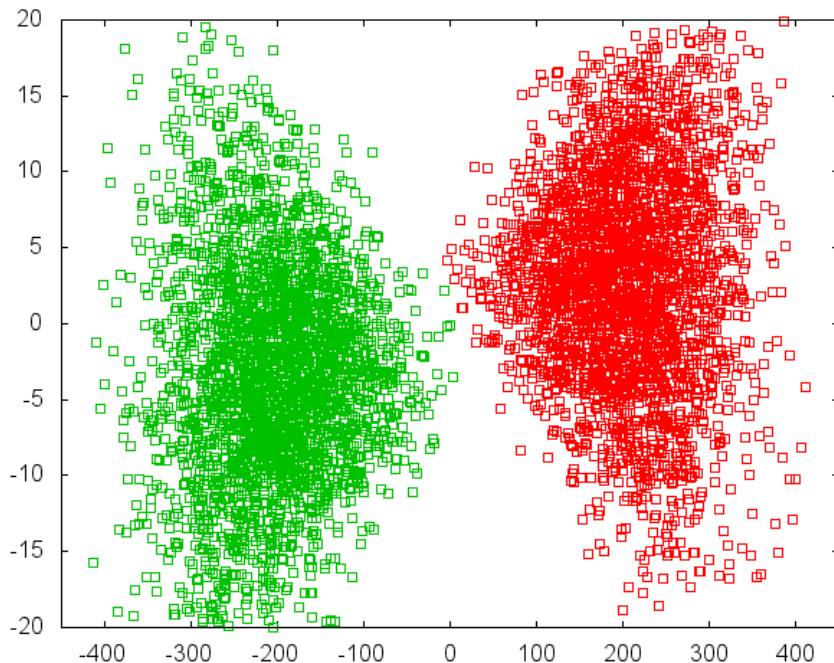
$$\frac{\mathcal{L}}{\mathcal{L}_0} = \exp\left(-\frac{\Delta y^2}{4\sigma_y^2}\right)$$

Luminosity and Offset

Luminosity loss for beam offsets depends strongly on disruption parameter

Can become a problem

CLIC at 380GeV



$$\frac{\mathcal{L}}{\mathcal{L}_0} = \exp\left(-\frac{\Delta y^2}{4\sigma_y^2}\right)$$

Beam-beam Deflection

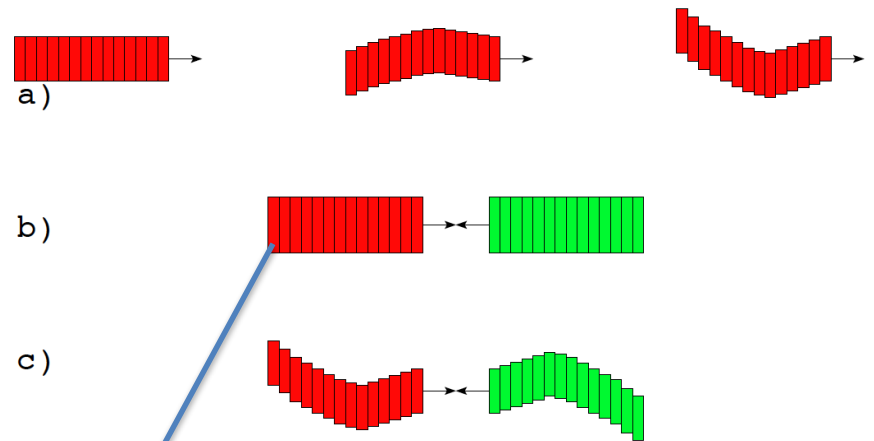
Keeping Beams in Collision

The Banana Effect

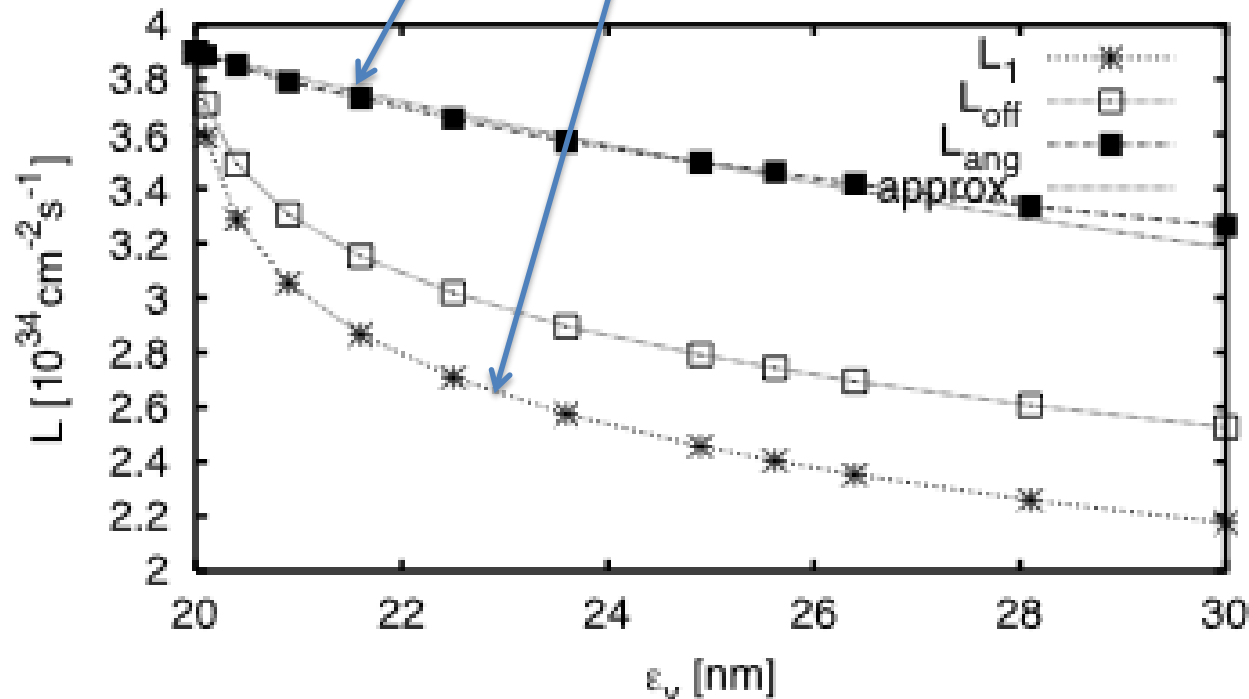
a) Wakefields+dispersion can create banana-shaped bunch in main linac

b) Do not model with projected emittance

c) The correct shape should be used



For large disruption
banana can reduce
luminosity



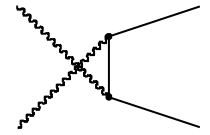
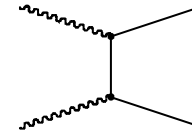
Study done for TESLA
Similar disruption as ILC

Electron-Positron Pair Production

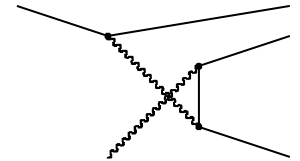
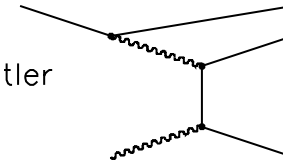
Colliding photons can produce electron-positron pairs (**incoherent pair production**)

$O(10^5)$ per bunch crossing

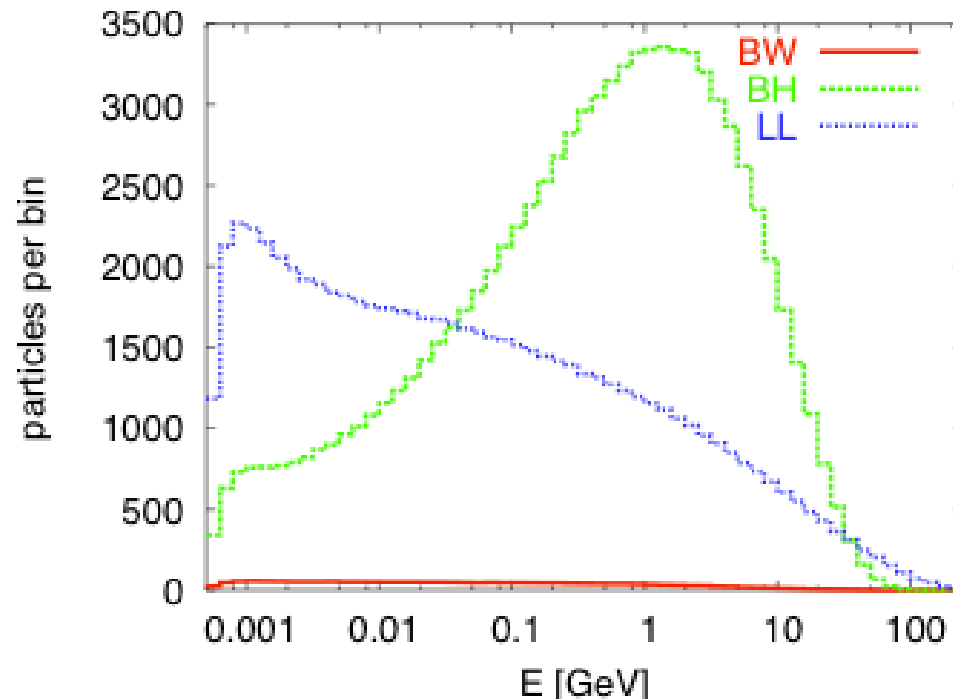
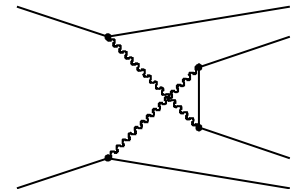
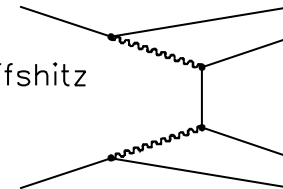
Breit–Wheeler process



Bethe–Heitler process



Landau–Lifshitz process



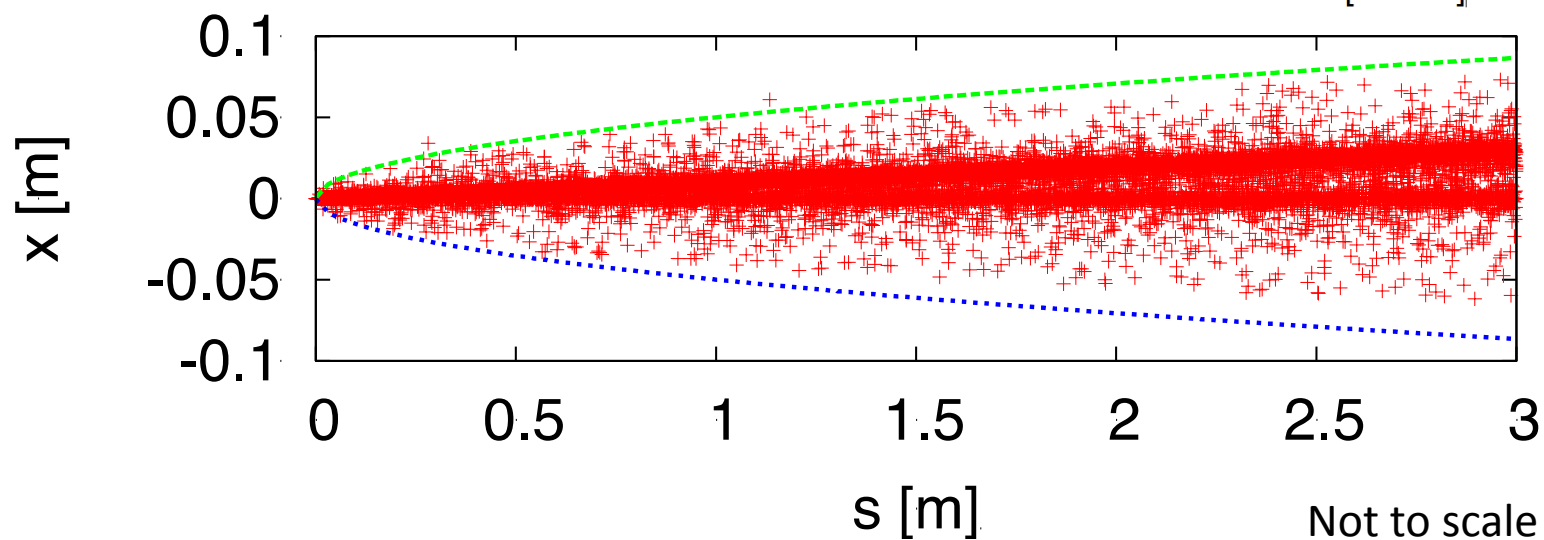
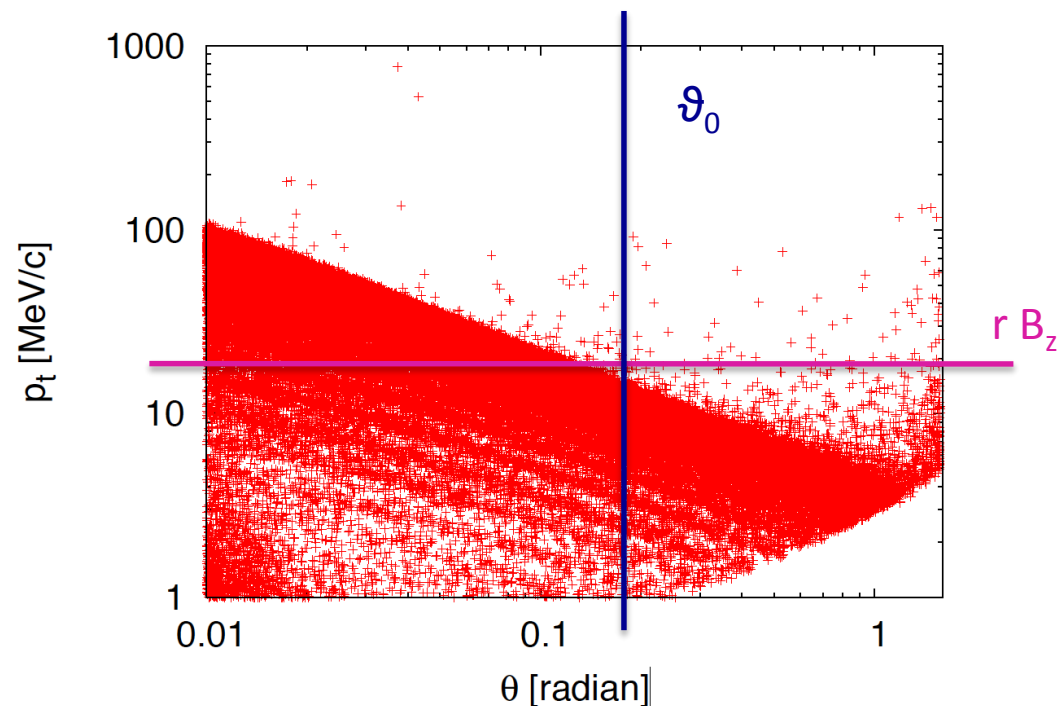
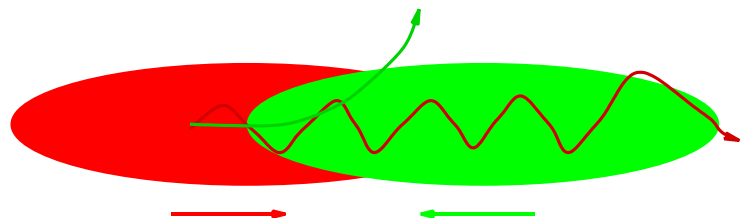
Note: beamsizes effect gives some reduction, ignored here

Also muons can be produced, but small rate

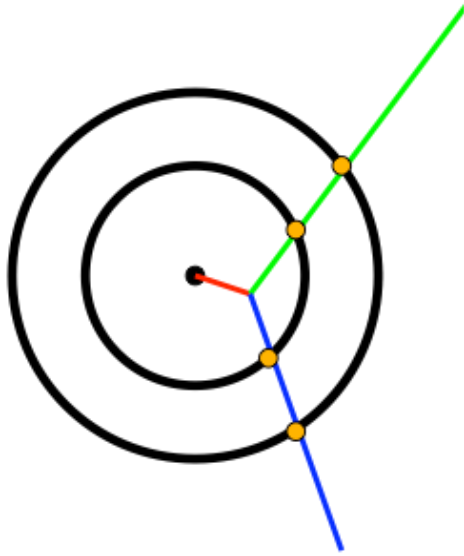
ILC and CLIC Main Parameters

Parameter	Symbol [unit]	ILC	CLIC	CLIC
Centre of mass energy	E_{cm} [GeV]	500	380	3000
Total luminosity	L [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	1.8	1.5	6
Luminosity in peak	$L_{0.01}$ [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	1	0.9	2
Particles per bunch	N [10^9]	20	5.2	3.72
Bunch length	σ_z [μm]	300	70	44
Collision beam size	$\sigma_{x,y}$ [nm/nm]	474/5.9	149/2.9	40/1
Vertical emittance	$\varepsilon_{x,y}$ [nm]	35	40	20
Photons per beam particle	n_γ	1.9	1.5	2.1
Average photon energy	$\langle E_\gamma/E_0 \rangle$ [%]	2.4	4.5	13
Coherent pairs	N_{coh}	-	-	6.8×10^8
Their energy	E_{coh} [TeV]	-	-	2.1×10^8
Incoherent pairs	N_{incoh}	196×10^3	58×10^3	300×10^3
Their energy	E_{incoh} [TeV]	484	187	2.3×10^4

Incoherent Pairs

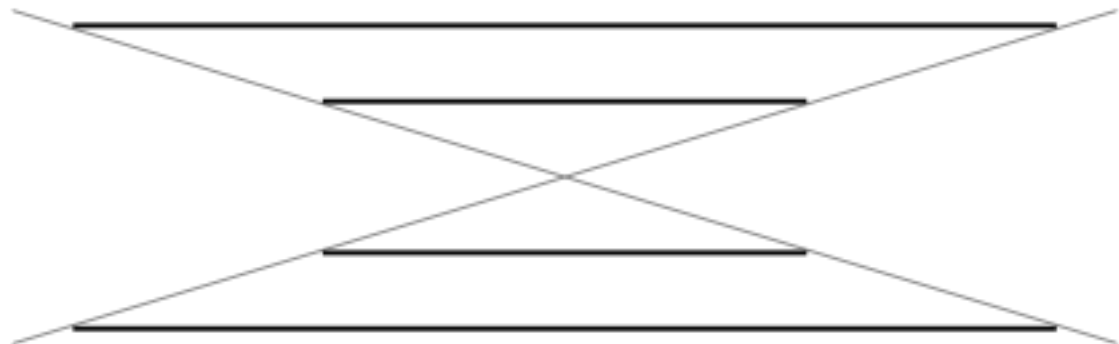
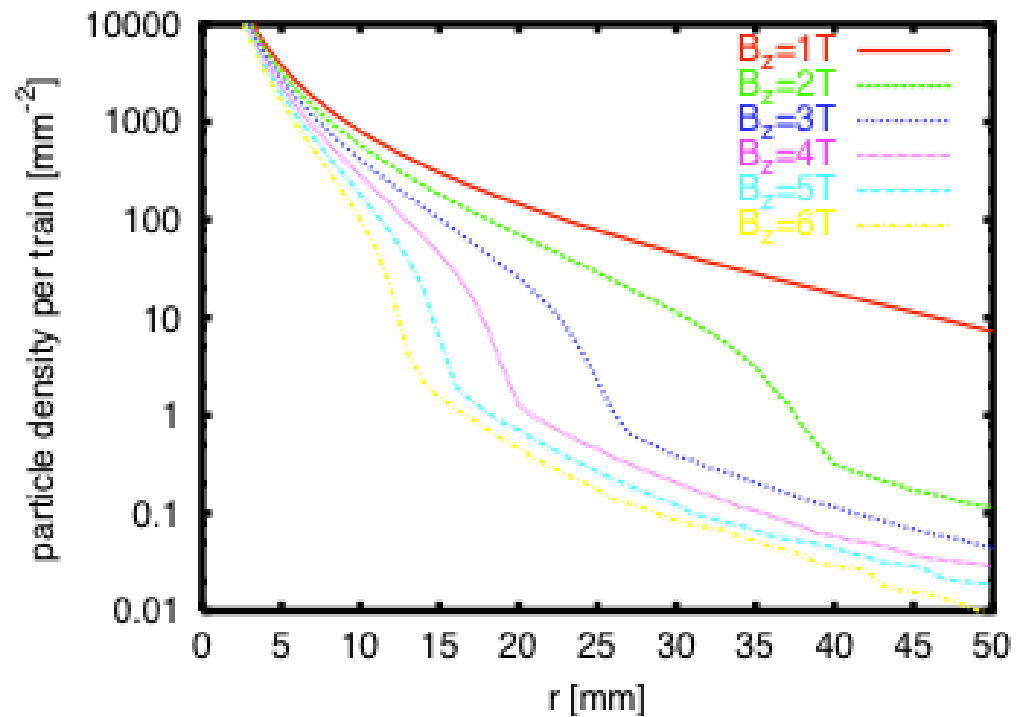


Vertex Detector



To identify decays

Need a certain angular coverage
Need a minimum radius to avoid
pairs



Conclusion

- Beam-beam effects limit the luminosity in linear colliders
 - Exact limitation depends on the physics run
 - Flat beams maximise luminosity
 - Non-linear QED can appear
- Machine background poses important constraints on the experiment

Reserve

Classical Regime

Beamstrahlung is like synchrotron radiation

$$\Upsilon \ll 1$$

Number of photons is

$$n_\gamma \propto \Upsilon \frac{\sigma_z}{\gamma} \propto \frac{N}{\sigma_x + \sigma_y}$$

Energy of photons is

$$E_\gamma \propto \Upsilon E_0 \propto \frac{N}{\sigma_x + \sigma_y} \frac{\gamma}{\sigma_z} E_0$$

Useful approximate formulae averaged over the beam

$$\langle n_\gamma \rangle \approx 2.12 \frac{\alpha r_e N}{\sigma_x + \sigma_y}$$

$$\langle E_\gamma \rangle \approx 0.462 \Upsilon E_0$$

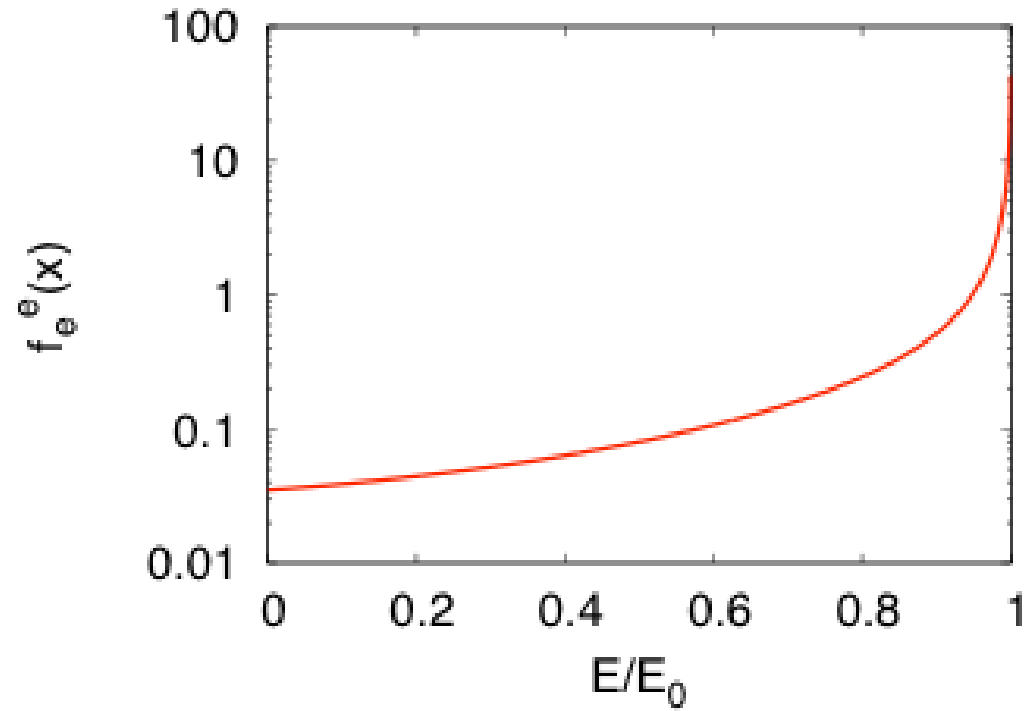
Initial State Radiation

Electrons often emit a photon before the collision

⇒ Initial State Radiation

The electron can be replaced by a spectrum of electrons

This yields a luminosity spectrum



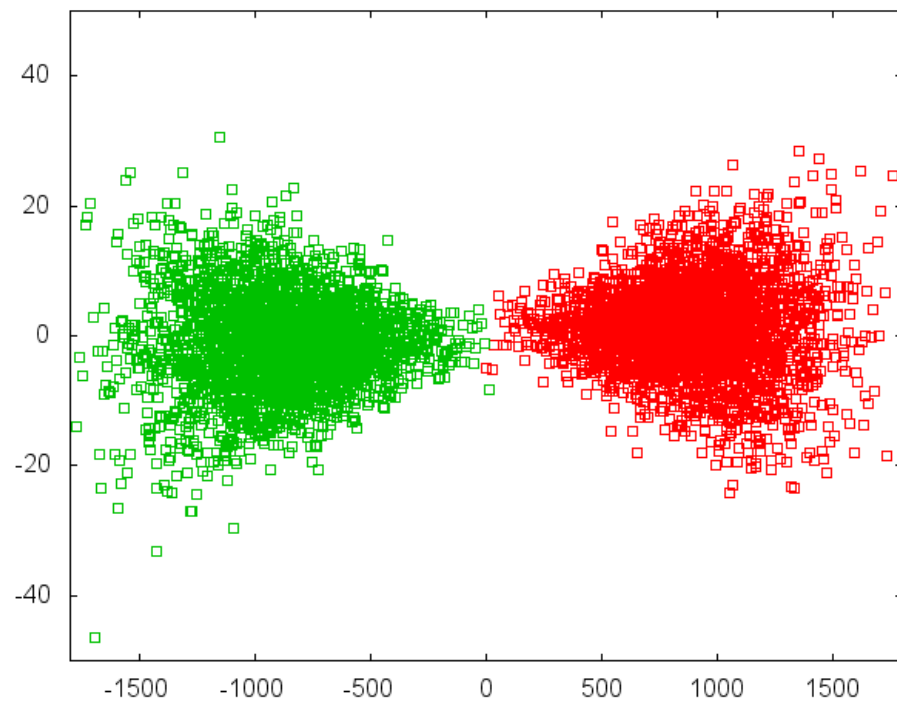
If you really want some formulae:
This is what PYTHIA uses

$$f_e^e(x, Q^2) = \frac{\beta}{2}(1-x)^{(\frac{\beta}{2}-1)} \left(1 + \frac{3}{8}\beta\right) - \frac{\beta}{4}(1+x)$$
$$\beta = \frac{2\alpha}{\pi} \left(\ln \frac{Q^2}{m^2} - 1 \right)$$

$$Q^2 = s = 4E_{cm}^2$$

ILC and CLIC Main Parameters

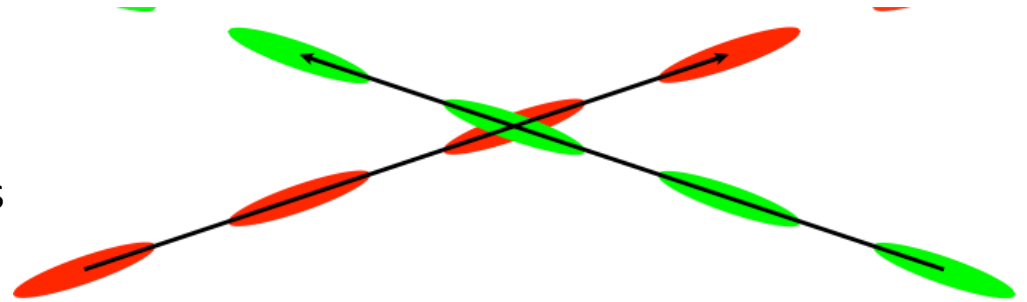
Parameter	Symbol [unit]	SLC	ILC	CLIC	CLIC
Centre of mass energy	E_{cm} [GeV]	92	500	380	3000
Total luminosity	L [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	0.0003	1.8	1.5	6
Luminosity in peak	$L_{0.01}$ [$10^{34}\text{cm}^{-2}\text{s}^{-1}$]	0.0003	1	0.9	2
Particles per bunch	N [10^9]	37	20	5.2	3.72
Bunch length	σ_z [μm]	1000	300	70	44
Collision beam size	$\sigma_{x,y}$ [nm/nm]	1700/600	474/5.9	143/3.3	40/1
Vertical emittance	$\epsilon_{x,y}$ [nm]	3000	35	40	20
Horizontal disruption	D_x	0.6	0.3	0.26	0.2
Vertical disruption	D_y	1.7	24.3	11.3	7.6



Note: Crossing Angle

Have crossing angles

- ILC: 14mradian
- CLIC: 20mradian
- to reduce effects of parasitic crossings
- to extract the spent beam cleanly

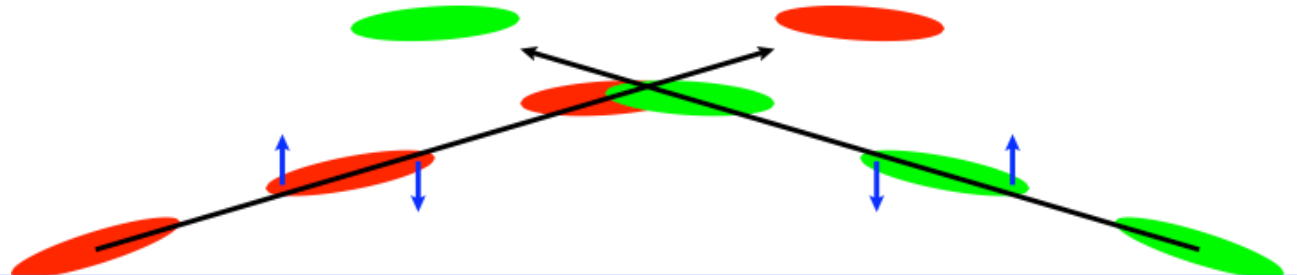


Luminosity with crossing angle

$$\mathcal{L} = H_D \frac{N^2 f_r n_b}{4\pi \sigma_x \sigma_y} \frac{1}{\sqrt{1 + \left(\frac{\sigma_z}{\sigma_x} \tan \frac{\theta_c}{2} \right)^2}}$$

0.1-0.2

Use crab cavities:

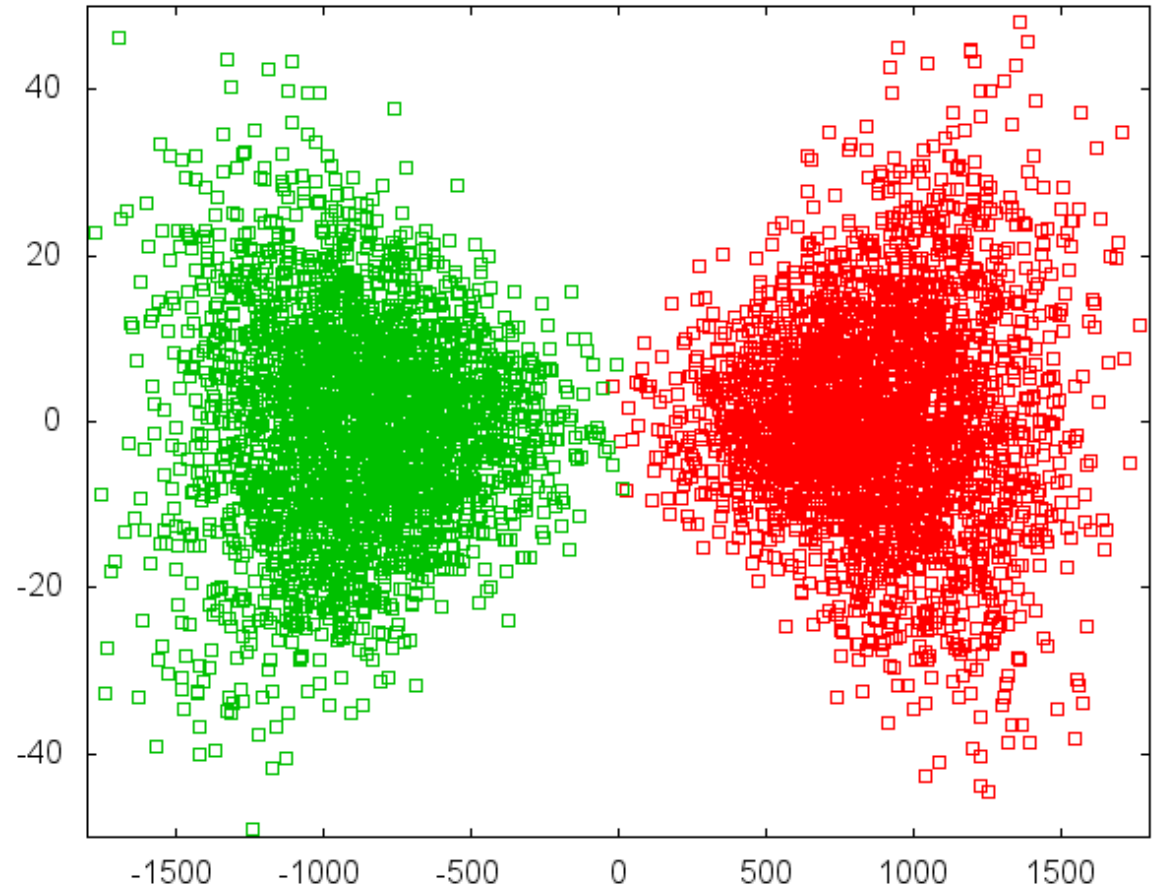


Can ignore crossing angle for now

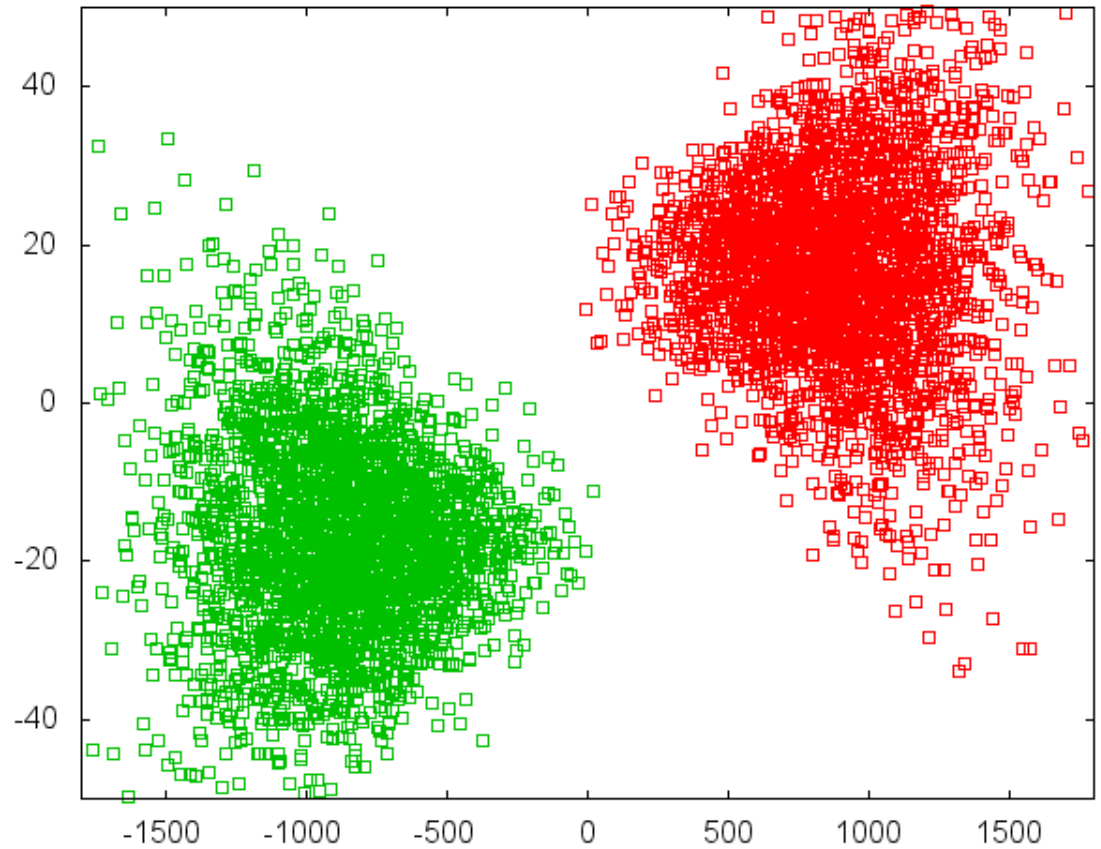
ILC and CLIC Main Parameters

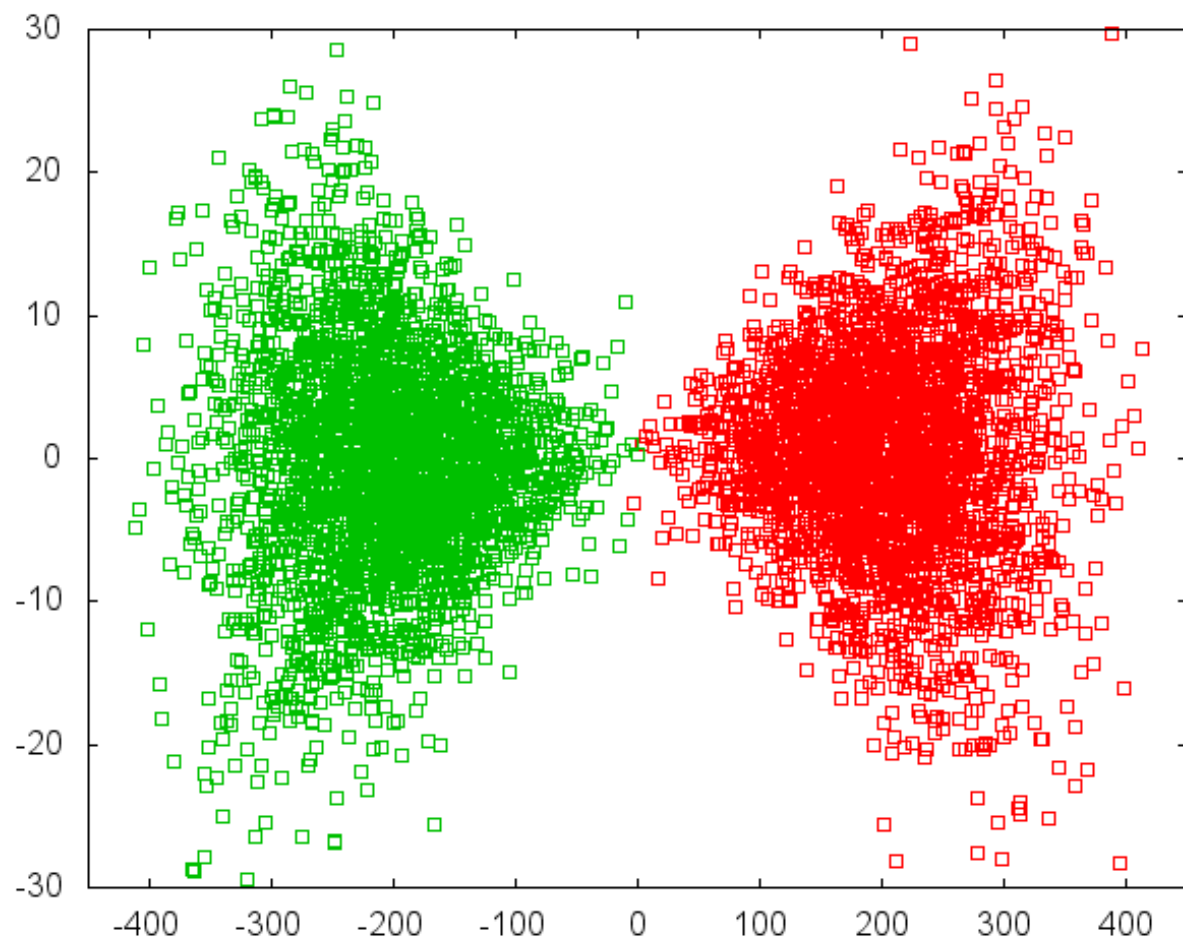
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Beam-beam



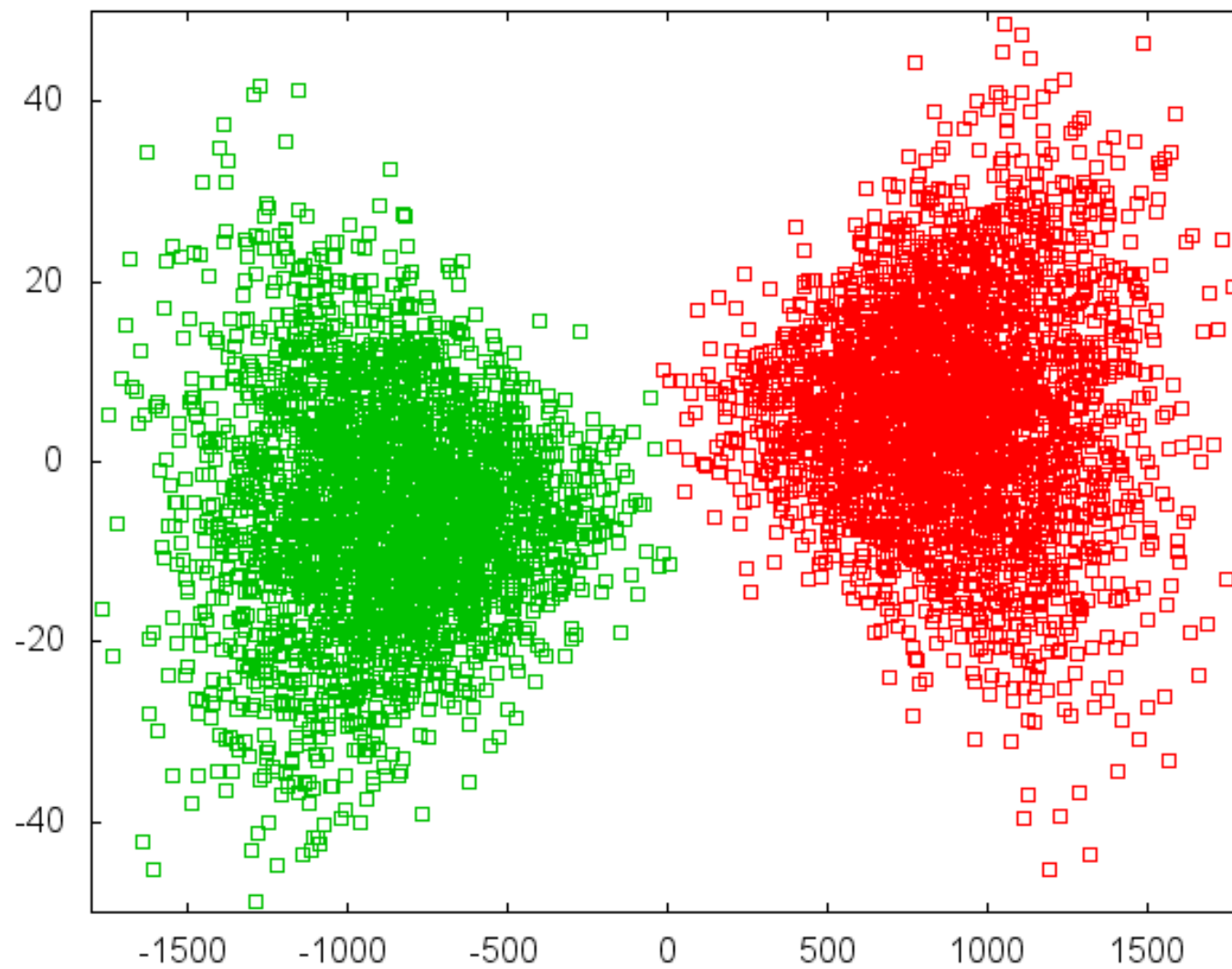
Beam-beam Interaction



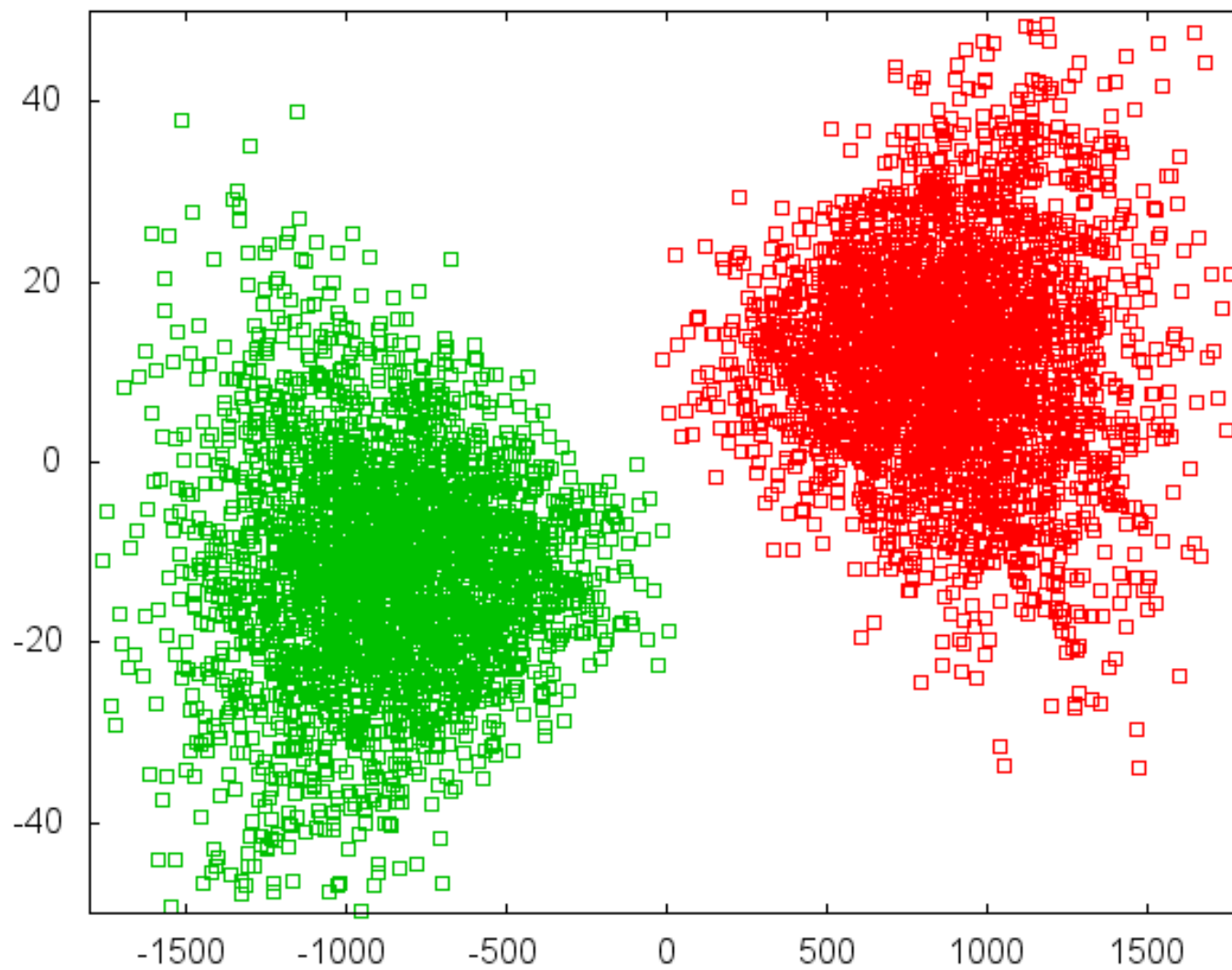


The Banana Effect II

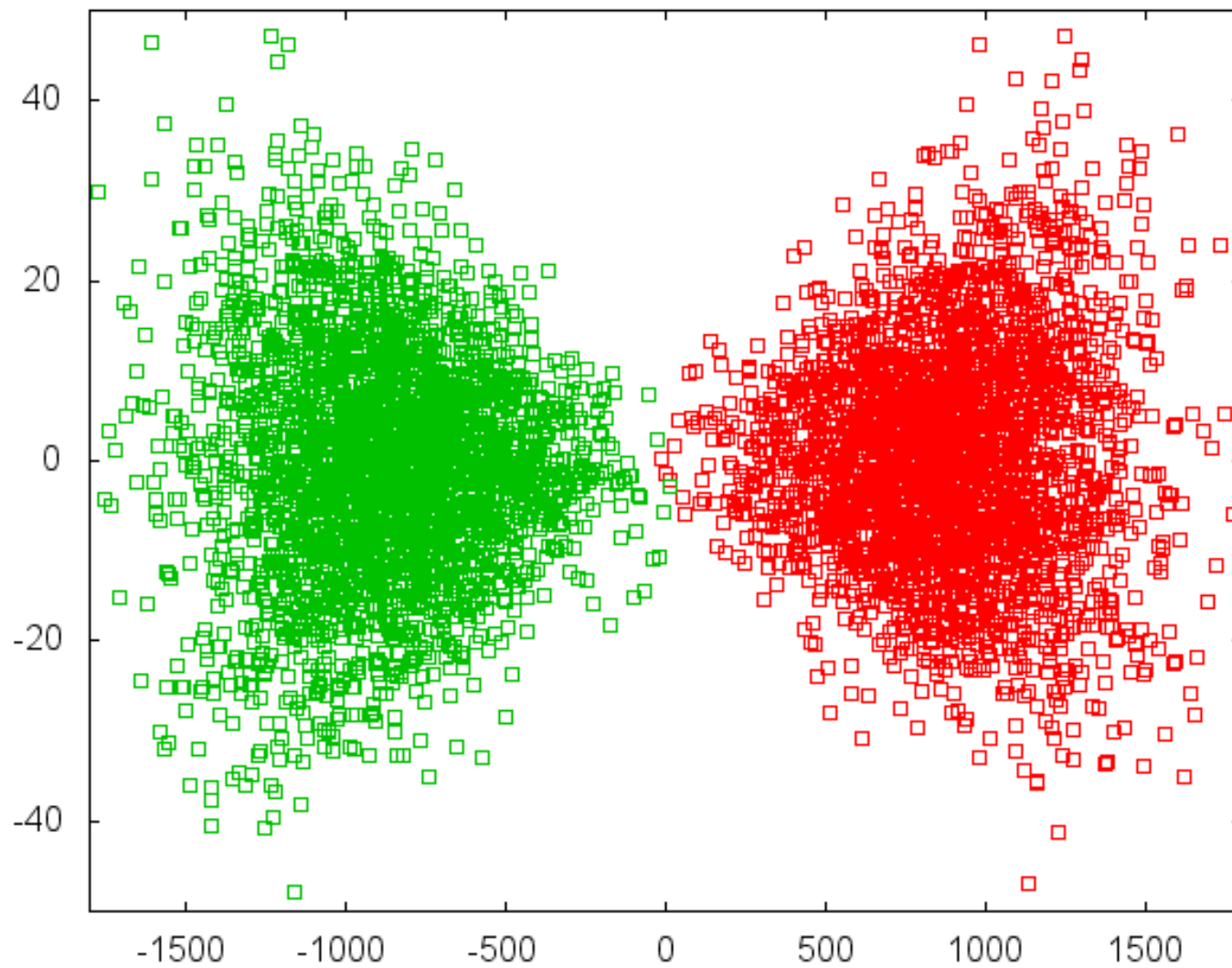
ILC



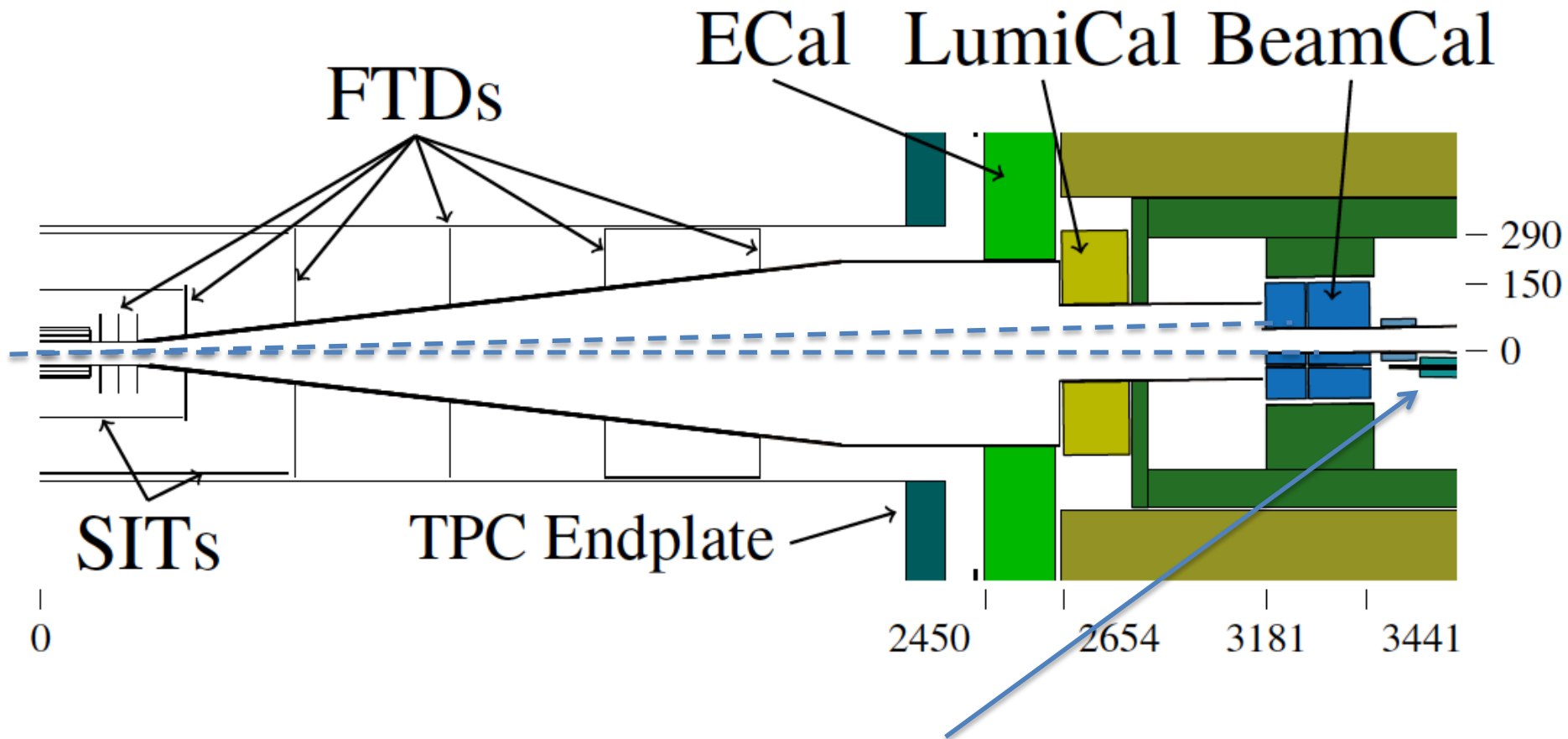
ILC II



The Banana Effect II



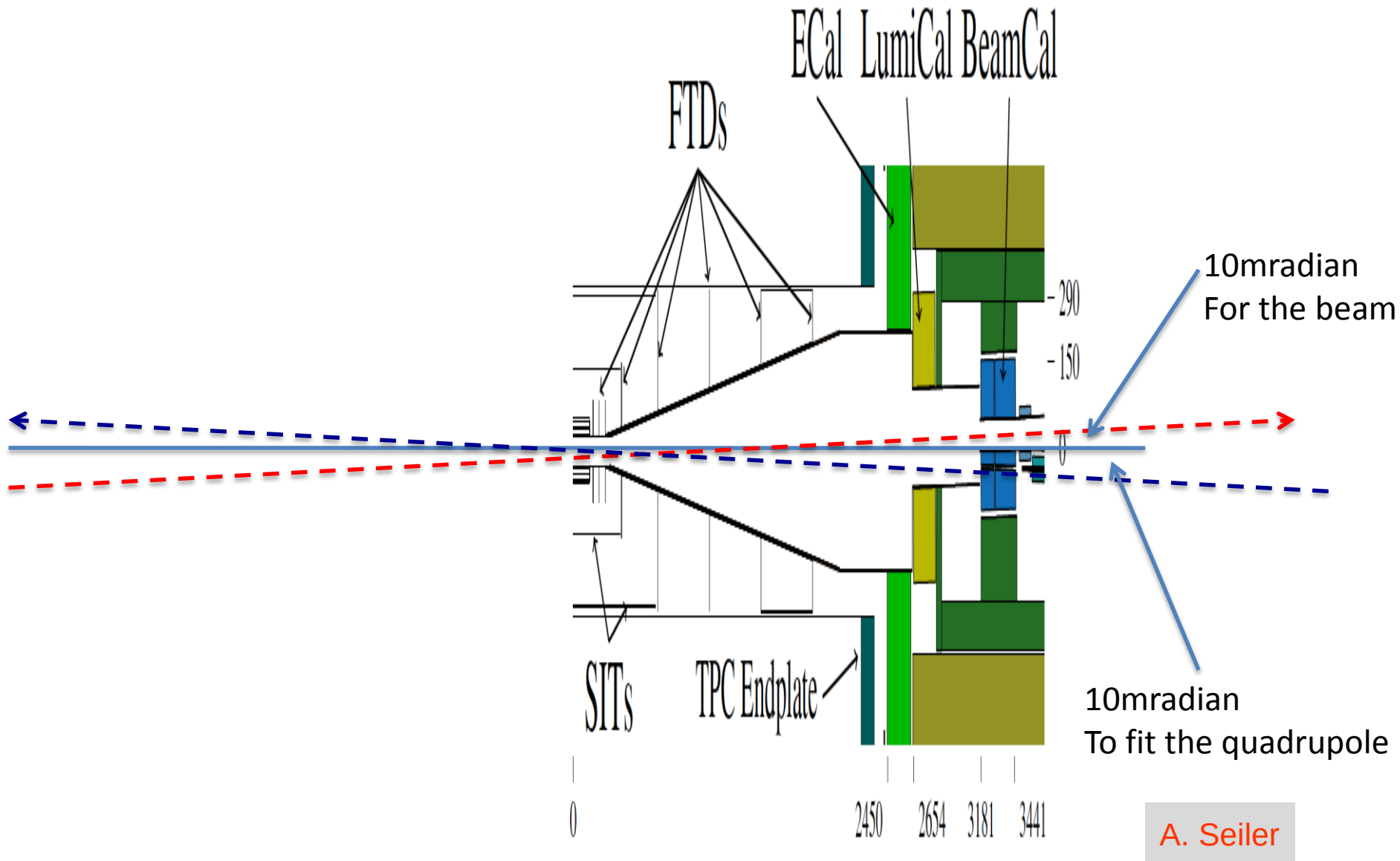
CLIC Inner Detector Layout



The last focusing magnet of the machine is inside of the detector

A. Seiler

Inner Detector Layout



Parasitic Crossings